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Assessing impacts of introducing ship-to-store service on sales and returns in omnichannel retailing: A data analytics study

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ABSTRACT

Omnichannel retailing features, such as ship-to-store (STS) service, are designed to deliver a seamless shopping experience for customers. For a retailer, introducing omnichannel capabilities requires major investments to integrate physical stores and online marketplaces, yet holds a promise of potentially enhancing revenue streams from both brick-and-mortar (BM) stores and online store channels. We assess the promise of ship-to-store capabilities by analyzing transactional data from a national jewelry retailer to study impacts of introducing ship-to-store on a retailer's operating performance, in terms of sales and customer returns. Contrary to expectations, the findings show that online sales decreased after ship-to-store was introduced, although BM store sales increased. Detailed analysis of the transactional data suggests that, after STS implementation, some customers switched from the online channel to the brick-and-mortar channel. This switch occurred mainly for high-value purchases. The customers who actually remained with and fully completed a sale using the ship-to-store service typically were those that bought low-value items. Our findings also suggest that introducing ship-to-store increased cross-channel customer returns of online purchases to physical stores. Concurrently, these new ship-to-store returns generated additional BM store sales. The paper contributes by showing how introducing ship-to-store service can have different impacts in terms of sales and returns across a retailer's channels.

1. Introduction

We assess impacts of introducing *ship-to-store* service on retailer sales and customer returns across multiple retailing channels. With the advent of technology-enabled shopping alternatives, retailers began augmenting their retailing channels with many new service processes, the collection of which has evolved into what today is known as *omnichannel retailing*. The main focus of omnichannel retailing is to offer consumers a seamless shopping experience, no matter which channel they use (Rigby, 2011; Brynjolfsson et al., 2013; Bell et al., 2014). With omnichannel retailing, customers can buy online, buy in stores, or buy via several other shopping modes (e.g., catalogs, mobile devices). Among the many models of omnichannel purchasing and order fulfillment, major retailers today offer *buy-online and pick-up-in-store*, *ship-to-store*, *ship-from-store*, and *reserve-online and pick-up-in-store* services to meet customer expectations.

Many omnichannel service processes are designed to draw customers into physical stores (RIS, 2012; MA, 2014) and thereby increase store traffic (Yantra, 2005; Lieb, 2015). Retailers do this via purchase options such as *ship-to-store* or *buy-online and pick-up-in-store*, as well as via return options such as *buy-online-return- to-store* (Zhang et al., 2010). Store traffic is essential to increase sales (Gulati and Garino, 2000; Bell et al., 2014), either through impulse purchases or through the assistance of store employees (Fisher and Raman, 2010; Mani et al., 2015). For example, at the national jewelry retailer we study, employee guidelines and training materials indicate that ship-to-store and buy-online-return-to-store events are important selling opportunities. For ship-to-store, sales associates are directed to use the occasion to cross-sell accessories and attendant items, along with profitable services like warranties. For returns, the selling prescription for the salesperson is to convert the return into an exchange or to up-sell to a more expensive item.

From a customer's perspective, the benefit of omnichannel integration is an increase in the value proposition offered by retailers (Gallino and Moreno, 2014; Gao and Su, 2017a) due to lower transaction costs, higher service quality, and lower perceived risk (Herhausen et al., 2015). With ship-to-store service, another customer benefit is a perceived increase in product variety because retailers can augment physical store inventory with virtual inventory offered on the Internet (Radial, 2016).

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The envisioned benefits of omnichannel retailing for retailers and customers have not been lost on practitioners, as evidenced by the sheer number of retailers pursuing omnichannel strategies. Even so, retail executives still worry about introducing effective omnichannel processes. To date, little academic research has studied the efficacy of omnichannel retailing tactics to stimulate demand, drive store traffic, or enable customer returns, which we address.

While omnichannel retailing provides benefits to customers, and ostensibly to retailers, implementation by necessity involves the adoption of costly and difficult-to-implement information and material handling technologies that can generate new operational challenges (Davis, 2008; Zhang et al., 2010). Omnichannel retailing requires integrating promotion campaigns, assortment planning for online and offline channels, inventory systems, and warehouses (Gallino and Moreno, 2014). It also can require multi-channel order management systems, integration to third-party partners, and many other internal or outsourced retailing systems (Perdikaki et al., 2015). Historically, retailers have had a hard enough time accurately tracking their store inventory in the first place (DeHoratius and Raman, 2008), let alone having the capability to offer inventory visibility across multiple channels in a real-time manner. As a case in point, while 60% of retailers in a recent survey claim they have implemented inventory visibility across channels, 80% of them report that their systems need improvement due to implementation issues (BRP, 2016). Thus, there is a clear tradeoff between the costs and challenges associated with implementing omnichannel retailing and the benefits that may arise from such systems. Complicating matters is that there is a variety of omnichannel process alternatives, each having different operational complexities and distinctive value propositions.

This paper focuses on implications of introducing *ship-to-store* service. Although some people may use the terms *ship-to-store* (STS) and *buy-online and pick-up-in-store* (BOPS) interchangeably, the two are in fact largely different service processes with different fulfillment tradeoffs (Acimovic and Graves, 2015). In short, BOPS provides customers with real-time store-level product availability information, lets customers complete transactions online, and allows customers very soon thereafter to pick up the items in a store at their convenience (Gao and Su, 2017a). BOPS reduces shopping transaction costs for customers since items are picked and packed by store employees prior to customer pick-up. In contrast, with STS, customers complete a purchase transaction online, and then wait for a notification about delivery of purchased items to their local stores, free of charge. With STS, shipping of items from a central distribution center (DC) generally occurs, even if the item is already available at the store, as is the case with the retailer we study. Hence, BOPS uses in-store inventory to fulfill customer demand, while STS uses centralized fulfillment.

In practice, we observe many variations of BOPS and STS processes, including hybrids of the two, with some involving local store-to-store inventory transshipment. Table 1 compares eight national retailers in terms of the BOPS or STS features that they offer to customers. Clearly, there is no single omnichannel strategy that retailers are pursuing. We thus are left to wonder why a retailer chooses to offer a variant of such services. Academic research adds little clarity. We are aware of only two research works that address BOPS (Gallino and Moreno, 2014; Gao and Su, 2017a) and only one that addresses STS (Gallino et al., 2017). Clearly, this gap in the literature, compared against the extensive offerings of these omnichannel services in the marketplace, signals a significant research opportunity.

Table 1
BOPS vs. STS.

Firms	Offers BOPS	BOPS lead time	Offers STS	STS lead time
Walmart	Yes	4 h	Yes	7-10 business days
Best Buy	Yes	45 min	Yes	3-7 business days
Lowe's	Yes	20 min	No	–
Kohl's	Yes	4 h	No	–
Macy's	Yes	4 h	No	–
REI	No	–	Yes	7-10 business days
Michaels	No	–	Yes	5-7 business days
Kirkland's	No	–	Yes	7-10 business days

In this paper, we empirically investigate the impact of introducing STS service on a retailer's operations by using a series of quasi experiments. We use the difference-in-difference (DID) econometric methodology to compare the pre and post periods of STS introduction. To do so, we collected a proprietary data set from a national jewelry retailer that implemented STS service. As with virtually all fashion items, jewelry is an experience good, and because of this, there is a significant risk of an item in this product category getting returned, since the customer is only able to assess its quality and fit after it is received.

The data set in its entirety spans four years with more than 20 million customer-level purchase and return transactions. The jewelry retailer operates more than 1000 stores in the U.S. and Canada. Using a subset of this data set that corresponds to a two-year time window (one year before and one year after) surrounding the point in time of the STS introduction, we are able to observe how purchase and return activities of customers change after the retailer introduced the new STS service and how these changes affect retailer performance.

Given the nascent state of research, we are positioned to make several contributions. Building on firm-level transaction cost theory and consumer utility maximization, we develop research hypotheses regarding the STS service. Theoretical reasoning leads us to hypothesize that both online and brick-and-mortar (BM) channel sales should increase after introduction of STS. Overlapping the findings of Gallino and Moreno (2014) for BOPS in furniture/housewares retailing, we do find that BM store sales increase after the jewelry retailer's STS rollout, while at the same time online sales actually decline post-STS. Moreover, the increase in BM sales is larger than the decrease in online sales. Plainly, there is more to the story than a simple channel shift of demand. A key point to note as well is that the theoretical underpinning for an increase in BM store sales resides with an increase in online sales. Ostensibly, the new availability of STS should directly stimulate online demand, and through the process of store pick-ups, generate store traffic and hence store sales. Yet, we find that online sales *decline*. Clearly, with this retailer's introduction of STS, a more nuanced explanation is needed.

In short, after digging into details within customer transaction data, we find that while customers may be drawn via STS service to make purchases online, many customers subsequently decide not to wait for store delivery, and instead opt to go directly to a store and immediately buy similar merchandise. This activity occurs mainly for high-value items. What is striking here is that there is no way for customers to know with certainty that the specific items they are interested in are even available at the store, since store inventory information is not made available online. In fact, only roughly half of the items available via STS are also available in stores, so long as they are not out-of-stock. We speculate that the research that customers conduct online makes them comfortable enough to believe that the selection offered in local stores will satisfy their needs, whether or not the online item is immediately available. In contrast, the online customers who do end up using STS are customers that mainly purchase low value items, for which the relative savings in shipping cost afforded by STS is disproportionately greater. This finding contributes by showing how a new STS service can lead to non-uniform customer actions across retail channels.

Another facet regarding STS introduction is its effect on product returns. We find that cross-channel customer returns increase. We also find that product returns of sales made at BM stores decrease, while returns for online purchases remain unchanged. Customers who switch from pure online shopping to STS service via the BM channel have conducted prior research online, thus they should be more knowledgeable about their purchases compared to regular BM customers. This is known as *reverse showrooming* or *webrooming* (Bell et al., 2014; Verhoef et al., 2015), which we expect is a stimulus for the increase in BM store sales that we observe. Yet, the same phenomenon should also affect product returns. With increased online shopping motivated by introduction of STS, customers who enter a BM store are now arguably more knowledgeable about their own likes and preferences as well as about a product's ability to meet their needs. This finding contributes to the literature new insights into how the introduction of STS services can change the dynamics of consumer returns.

The paper is organized as follows. In §2, we review literature. In §3, we introduce research hypotheses. In §4, we introduce and test our models. Given our initial results, we provide an extended analysis in §5. Finally, we provide managerial insights, offer future research suggestions, and conclude in §6.

2. Literature review

Our research resides at the Marketing-Operations interface, at the confluence of three streams of literature. The first stream investigates channel integration and omnichannel retailing. The second stream examines channel switching behavior of customers. The third stream addresses consumer returns in the context of omnichannel retailing. We discuss each stream and position our work with respect to them.

Research on channel integration and omnichannel retailing only began to attract scholarly attention recently. Herhausen et al. (2015) study the impact of channel integration on customer behavior and find that it increases the perceived service quality and decreases the perceived risk inherent to online shopping. Cao and Li (2015) investigate the association between channel integration and firm performance and find that it stimulates sales growth depending on a firm's previous online experience and physical store presence. Brynjolfsson et al. (2009) examine cross-channel competition and find that online retailers face a higher level of competition from BM retailers for mainstream products relative to niche products. Huang and Van Mieghem (2014) empirically examine a setting where a firm operates the online channel as a showroom (catalog) and accepts orders at the BM channel. They evaluate the value of using online channel clickstream information to improve the BM channel operations and show that clickstream data can be used to predict the ordering probability, amount, and timing of BM channel orders. Bendoly et al. (2005) explore the association between product availability and consumer retention and show that in the case of an availability failure (out-of-stock), perceived channel integration can prevent customers from shopping at competing firms by attracting them to the alternate channel. Gallino et al. (2017) study the impact of introducing omnichannel functionalities on a furniture and housewares retailer's inventory decisions and show that STS service increases the contribution of the lowest-selling products to total sales (i.e., sales dispersion). Although these papers explore the overall performance impact of channel integration and cross-channel competition, none of them separately evaluates the impact of channel integration on the online and BM channels in terms of both sales and returns, as we do in this study. Furthermore, we also contribute to the literature by exploring how channel integration influences cross-channel returns.

In the second stream, several studies focus on channel switching behavior of customers after the existing channels are integrated. Ansari et al. (2008) explore customer channel migration and show that marketing efforts influence customers to shift to the online channel and increase their sales volume. Chintagunta et al. (2012) investigate the process of choosing a channel for grocery purchases and empirically show that relative transaction costs play important roles when consumers select between online and offline channels. Jareth et al. (2015) study the behavior of customers in a multichannel service environment and show that customers use different channels with different objectives. The authors find that customers use a telephone for important health-related concerns while they prefer the Internet for more structured information needs. Forman et al. (2009) study the competition between the online and offline channels and find that customers switch from online to BM channel after a store opens in a local community. In related work, Avery et al. (2012) investigate the impact of store openings on direct channels like the Internet and catalog sales. They show that the presence of a physical store decreases catalog sales, but does not affect Internet sales in the short run. In the long run, both catalog and Internet channel sales increase.

In another study, Bell et al. (2018) explore the impact of information provision on the shopping behavior of consumers and show that some customers switch from online to the BM channel with the introduction of a display-only showroom. Gao and Su (2017a) investigate the impact of BOPS service on store operations and find that offering BOPS may increase a retailer's customer base while making some existing customers shift from the online channel to physical stores. Similarly, Gallino and Moreno (2014) empirically explore the impact of offering BOPS service on a furniture and housewares retailer's online and physical store sales and show that BOPS increases store sales and reduces online sales. Note that these three papers identify customer channel switching as a result of providing in-store product availability information to online customers. With STS service, however, such information is not available to online customers. Hence, we contribute to the literature by exploring how online and BM customers respond to channel integration via STS service, in a situation where store product availability information is not provided.

The third stream focuses on consumer returns, another inseparable aspect of retailing that accounted for \$351 billion in returned revenues in 2017 (Appriss Retail, 2017). In this stream, Wood (2001) examines consumer returns in a remote purchase environment and shows that a lenient return policy increases the probability of an order. De et al. (2013) empirically investigate the impact of implementing certain web technologies on online product returns. Studying consumer returns in a multichannel context, Ofek et al. (2011) show how store assistance and pricing decisions change after introducing an online channel. Gao and Su (2017b) analyze three omnichannel information mechanisms—physical showrooms, virtual showrooms, and information availability—and find virtual showrooms may increase online channel returns if they lead to excessive customer migration from BM stores to the online channel. Griffis et al. (2012) empirically explore the returns of Internet consumers and find that prior customer experience with returning positively affects future repurchase behavior. Rabinovich et al. (2011) explore virtual assortments offered via Internet retailing and show that retailers may benefit from increasing their online assortment if they can successfully manage the increase in product returns arising from execution errors and product mismatches.

Given that consumer shopping habits have evolved to include multiple channels and different activities in each of those channels, Peterson and Kumar (2009) examine impacts of cross-channel purchases on product returns. They find that when customers buy familiar products from new channels, they return fewer items, whereas when customers buy unfamiliar products from new channels, they return more items. They also find that the number of product returns is positively associated with a customer's future purchase behavior. In related work, Bower and Maxham III (2012) compare fee-based and free return shipping alternatives in online retailing and show that customers who pay for return shipping decrease their future spending while customers who enjoy free return shipping increase their future purchase spend.

There is also a growing body of work concerning customer returns for experience goods. Ertekin (2018) examines whether in-store customer service experience dimensions for a jewelry retailer affect subsequent product exchange behavior, return process satisfaction, and future repurchase activity. Ertekin et al. (2017) study the relationship between in-store customer shopping experience and subsequent product returns. Combining jewelry retailer transaction data with customer survey responses, they suggest retailers might mitigate the impact of unpleasant store ambience on returns by improving the competence levels of sales associates. Our work draws from the same corporate data set as both Ertekin (2018) and Ertekin et al. (2017), although we address fundamentally different research questions. Ketzenberg et al. (2018) examine luxury department store data in an attempt to classify, identify, and predict behaviors of abusive opportunistic returners. We, too, predicate our analysis on customer transaction data. Compared to these studies that focus on consumer perceptions and behaviors, we contribute by investigating the change in returns that arises from a retailer introducing a free STS option for purchases, which effectively increases return policy leniency by eliminating hassles associated with returning online purchases.

3. Research hypotheses

We draw in part upon theoretical perspectives of transaction cost economization by economic actors to develop research hypotheses. The transaction cost perspective argues that transaction costs of exchange determine how a firm will structure its business processes (Zaheer and Venkatraman, 1995). Building on human behavioral assumptions of *bounded rationality* and *opportunism*, transaction cost theory suggests an economic actor will attempt to reduce transaction costs to make commercial transactions more efficient for themselves (Williamson, 1975, 1985). Transaction cost literature proposes that a wide variety of financial and non-financial transaction costs associated with the exchange of goods between economic actors (i.e., sellers and buyers) can affect the organization of commerce, including explicit fees and shipping costs, and information costs involved in product search. Depending on how transaction costs are driven by the operant constructs of transaction cost theory – *uncertainty*, *asset specificity*, and *transaction frequency* – economic agents will choose to do business via either a market, a hierarchy (i.e., an internalized transaction), or an intermediate hybrid governance structure based on relational (i.e., social exchanges, collaborative) governance (Zaheer and Venkatraman, 1995).

Prior retailing research builds upon facets of *uncertainty*, *asset specificity*, and *transaction frequency*, each of which can drive transaction costs, which in turn may decrease the willingness of economic actors to execute commercial transactions. Transaction cost theory has been applied in OM/SCM retailing research on consumer returns (Griffis et al., 2012). Prior empirical work also applies transaction cost theory constructs to analyze consumer decisions regarding use of in-store versus online shopping (e.g., Grønhaug and Gilly (1991), Liang and Huang (1998), Teo et al. (2004), Teo and Yu (2005), Chircu and Mahajan (2006)). Arguably, the same tenets of transaction cost theory can be applied to consumers acting as economic agents making purchases, as in Griffis et al. (2012). Instead, however, we draw upon the Tyagi (2004) consumer utility maximization perspective to link firm-level transaction cost economization to customer-level utility implications of the introduction of the jewelry retailer's new STS offering. Tyagi (2004) shows that when a retailer's service is transformed by market-level technological developments (e.g., technology for STS) that drive lower consumer transaction costs, those lower transaction costs increase total demand.

To begin, we explain how STS should affect online channel sales and returns. Next, we motivate how STS should impact BM channel sales due to an expected increase in physical store traffic arising from STS in-store pick-ups. We then explain how the customer convenience of returning an online STS purchase to a BM store should increase customer returns, and subsequently, increase BM channel sales.

3.1. Impact of STS on the online channel

Jewelry purchase transactions are characterized by a substantial degree of service co-production involving both retailer personnel/systems and customers. That is, customers play a substantial role to influence both the quality and efficiency of the service delivery process. From a transaction cost perspective of *transaction frequency*, jewelry purchases, being experience goods, are for most consumers a fairly infrequent type of transaction event. For a retailer, this infrequency innately positions jewelry selling as lacking in service co-production efficiency, due to a lack of economies that would arise if consumers frequently bought jewelry. A lack of consumer *transaction frequency* creates a situation characterized by higher retailer sales process *uncertainty*, since the consumer does not regularly educate him/herself about jewelry or practice the process of buying jewelry (i.e., exposing salespersons and customers to *bounded rationality*). As such, a jewelry retailer may inadvertently or intentionally miscommunicate aspects of products offered for sale, while a consumer may inadvertently or intentionally miscommunicate needs and desires – both instances creating an environment for potential opportunism.

Jewelry retailer *uncertainty* about buyers' needs requires retailer personnel to devote more effort to facilitate such sales, subsequently also forcing consumers to devote more time and effort to work with sales-persons in order to purchase the right piece of jewelry. Personal time and effort, as a non-substitutable resource when devoted to selling (or buying) just the right piece of jewelry, exhibits high levels of *asset specificity*, since there is a huge opportunity cost of devoting any person's time to the task of jewelry selling/buying. Situations such as this characterized by high asset specificity have been found associated with a greater degree of use of quasi-integration relational governance structures by trading partners (Zaheer and Venkatraman, 1995). From a transaction cost perspective, a jewelry retailer will use expectations about these transaction costs to decide which shopping functions to internalize via new quasi-integrated process innovations such as STS. Yet, only if transaction costs will be substantially reduced via STS mechanisms, will a retailer choose to introduce and use STS to conduct sales transactions.

A key driver of transaction cost reduction via STS introduction involves getting the customer into the store. High intensity sales processes with significant information exchange, such as that with jewelry sales, is facilitated by interpersonal interactions. By definition, online STS customers must visit physical stores to pick up their merchandise. From a transaction cost perspective, for the retailer, the STS process reduces *asset specificity* related transaction costs inherent in getting customers into a store, in order to then demonstrate appropriate products for them. The STS process also reduces *purchase infrequency* derived transaction costs because it enables a multi-step cycle of purchase-to-return-to-purchase events, characterized by consumer need sharing, retailer information collection, customer product trials, elucidation of product misfit, and in-person opportunities for resolving of a product misfit.

In addition, the retailer we study absorbs the direct e-retailing channel cost of shipping the product from its warehouse to the customer for items over a threshold purchase value. STS should reduce these costs due to preexisting deliveries to stores that replenish store inventory. The incremental cost of adding STS purchases to these preexisting store deliveries will necessarily be substantially less costly than using third party logistics providers such as USPS or UPS. Moreover, consolidating delivery of store inventory with STS purchases will increases transaction frequency inside of the retailer's internal network, leading to lower transaction costs and economies of scale. Finally, STS reduces uncertainty because jewelry assets remain in the retailer's distribution system, in turn enabling better security and visibility of inventory.

Collectively, all of these elements reduce transaction costs. We proceed by developing our hypotheses for both online and BM channels by linking the reduction of retailer transaction costs through the introduction of STS to aggregate buyer behavior that is influenced by explicit transaction costs and a variety of risks.

3.1.1. STS and online sales

Online and offline shopping channels provide varying levels of order fulfillment services and forward/reverse retail distribution services that entail different levels of utility for customers. In a co-produced, quasi-integrated retailer-to-customer shopping transaction, operational mechanisms such as STS that reduce retailer transaction costs implicitly also should reduce consumer transaction costs of shopping. Retailer demand should increase with a reduction in these costs (Tyagi, 2004). We next describe theoretical and practical components making up this aggregate transaction cost reduction.

The theory of buyer behavior proposes a wide variety of inhibitors that can disrupt consumer shopping actions (Howard and Sheth, 1969). E-commerce research has also identified numerous transaction cost drivers that will inhibit consumers' online sales activity (Teo et al., 2004). From an explicit transaction cost perspective, we argue that requiring a high shipping fee is one inhibitor to online purchasing, since once a fee is paid, the fee cannot be allocated to alternate uses. Lewis et al. (2006) and Gumus et al. (2013) show that compared to shopping in a physical store, additional surcharges in online shopping, such as shipping and handling fees, can have a significant influence on purchase intentions of customers. Lewis (2006) empirically shows that shipping fees reduce online customer traffic and order sizes. Industry surveys also suggest a major reason why customers abandon online shopping carts is due to high shipping fees (Ernst and Young, 2001; UPS, 2015). Conversely, consistent with Bower and Maxham III (2012), 71 percent of consumers in an Ernst and Young survey responded that free shipping and delivery is a top reason for brand loyalty intentions (Ernst and Young, 2015). About 90 percent of consumers report that free shipping would make them shop online more frequently (Walker Sands, 2016). Hence, consistent with Kukar-Kinney and Close (2010), STS will increase online sales by eliminating shipping costs as a shopping inhibitor. While our retailer offers free shipping only for orders totaling over $149, should a different retailer offer free STS shipping without a minimum transaction size, then we would expect the impact of STS to be less. Relatively, free home shipping means less value-added for the STS option. As such, the higher the threshold for free shipping, the greater the influence of STS on online shopping behavior.

Secure home-delivery is another relevant implicit transaction cost that customers face that inhibits the purchase of goods online. Reolink (2017) argues that 23 million Americans' packages are stolen each year. Indeed, there are plenty of neighborhoods and delivery locations, representing tens of millions of persons in the US alone, in which it is simply not reasonable to leave packages unattended. Inner cities are obvious examples, but homes and apartments located in major metropolitan areas can pose particular problems with respect to receiving packages unattended. In such situations, many customers are obligated to wait at home for delivery or perhaps have packages delivered to a work location, both of which heighten delivery inconvenience. Introduction of quasi-integration via STS may reduce both the security risk and the inconvenience risk, wherein a store location provides an alternative, safe delivery option and in which a customer can pick up their items at a convenient time. Even Walgreens, when announcing ship-to-store service, makes clear that “with ship-to-store, customers have the ability to ship orders to their preferred Walgreens store if their residence or workplace isn't a secure option” (Walgreens Newsroom, 2016).

For a consumer, online shopping involves several risks, which include product performance risk, financial risk, and time/inconvenience risk (Danaher et al., 2003; Peck and Childers, 2003; Griffis et al., 2012). Each risk can be viewed as a driver of consumer transaction costs. Introduction of STS should reduce these risks. Ofek et al. (2011) note that risk associated with a product mismatch for a purchase is higher in an online channel than in a BM channel. This arises due to the lack of touch and feel available through online search. In the case of a product mismatch, consumers incur additional transaction costs to return merchandise. From this perspective, it is reasonable to expect that risk-averse customers are less likely to shop online (Forman et al., 2009). By implementing STS, the difficulty of returning an item is reduced, since customers may immediately make a hassle-free return at the time they pick-up their merchandise at stores. As such, reducing uncertainty should increase demand.

In summary, by introducing quasi-integrated STS technology developments, the retailer reduces many disparate consumer transaction costs and risks. In turn, these reductions should lead to increased overall demand (Tyagi, 2004).

Hypothesis-1a (H1A). The introduction of STS service will increase online sales.

Since the risk associated with leaving packages unattended using home delivery service increases with respect to the value of the purchased items, consumers may be discouraged from buying higher-priced products online. Of course, many retailers impose a threshold value limit, that if met, requires a customer signature on delivery, as does the retailer that we study (> $100). In fact, there even exist customers who ask for a signature requirement for their delivery from retailers, which originally do not require signature for their shipments (BestBuy-Forum, 2015; Apple-Forum, 2017). The signature requirement helps retailers to avoid customer complaints arising from lost or stolen packages. At the same time, the signature requirement heightens inconvenience risk associated with online purchase transactions because customers then must devote specific time and effort, losing any alternate use of that time, to wait at home for a package delivery.

From this perspective, when purchasing high value items, customers must either (1) accept the higher security risk associated with unattended home-delivery of items, (2) accept higher inconvenience risk associated with waiting at home for delivery, or (3) avoid purchasing high value items online. Again, STS by its very nature reduces both security risk and inconvenience risk for customers by allowing a store location to serve as a safe alternative delivery location and by allowing customers to pick-up merchandise at their convenience. Hence, we expect that any reduction in perceived risk will impact high-value items more than low-value items.

Hypothesis-1b (H1B). The increase in online sales due to introducing STS service will be greater for high-value products than for low-value products.

3.1.2. STS and customer returns of online sales

Another element of omnichannel retailing is the consumer flexibility offered by allowing cross-channel returns, such as the ability to buy online and subsequently make a return to a store. Consumer flexibility should lower the specific consumer effort/time of making a return, thereby lessening transaction costs of making returns, and thus, likely stimulating more returns. In a recent survey, 65% of consumers consider the ability to return an online purchase to a BM store as a positive feature of a hassle-free return experience (UPS, 2015). Although customers of the retailer that we study had a return-to-store (RTS) option before STS was implemented, we expect that introducing STS should also influence return behavior among online customers.

When STS customers pick up their merchandise in stores, they can immediately perform a physical fit of the product and resolve any remaining questions or concerns arising from the lack of touch-and-feel experience innate with online shopping. This *a priori* expectation further reduces transaction costs for the consumer and drives more commerce activity (Tyagi, 2004) by setting expectations of easy, hassle-free returns. Upon inspection, if customers find that an item does not match their needs, they can immediately make a simple and costless return while in the store. The marginal cost of making a return, given the customer is already at the store, is essentially zero. In effect, STS increases the perceived leniency of a return policy because customers do not need to repack the item, locate the receipt, create a return label for shipping, drive to the carrier (i.e., USPS or UPS) store to ship the item to an online returns center, or drive the item to the nearest BM store. Hassles are eliminated for the customer, which eliminates associated transaction costs. As a result of the perceived leniency of an easier return policy arising from STS, customers should feel more comfortable about the return process, which in turn will reduce price-type and psychological-type costs (see Chircu and Mahajan (2006)). Therefore, consistent with a consumer utility maximization perspective, with lower post-STS transaction costs of making returns, we propose that RTS incidents will increase after STS is introduced.

Hypothesis-2a (H2A). Introducing STS will increase RTS incidents.

Moreover, as we suggest above in H1A, after the introduction of STS service, online channel sales are expected to increase. Virtually by definition, an increase in online sales will increase returns given the probability of a mismatch, which is unaffected by the implementation of STS. Moreover, as indicated by H2A, since we expect RTS incidents to increase, we expect overall online returns to increase. For both reasons, we thus expect that overall customer returns for online purchases will increase.

Hypothesis-2b (H2B). Introducing STS will increase returns of online purchases.

3.2. Impact of STS on the BM channel

Store traffic is a necessary input for enabling in-store sales. Increasing store traffic and converting the increased traffic into new sales are vital to retailers (Perdikaki et al., 2012). Prior empirical studies show that sales generated in stores depend on both store traffic and labor (Lam et al., 1998; Mani et al., 2015; Chuang et al., 2016). Store traffic is critical for generating retail sales from several perspectives including impulse buying (unplanned purchasing) and directed sales initiatives. Impulse purchases account for between 27% and 62% percent of total sales at department stores (Wirtz, 2010; Bae et al., 2011), while 47 percent of BM channel customers engage in impulse purchases (eMarketer, 2015).

By definition, online STS customers must visit physical stores to pick up their merchandise. As previously discussed, surveys suggest customers that engage in BOPS or STS services end up spending more on additional items while they visit physical stores to pick up their online merchandise (eMarketer, 2015). According to the STS operating procedures of the retailer we study, sales associates are instructed to try to sell additional merchandise and warranty services at the time of pick-up. While there clearly are fixed costs of implementing STS for a retailer, once an STS-using customer enters the BM store, the retailer's marginal cost of acquiring the customer to sell items in person is essentially zero. Alternately, from a consumer utility perspective, once on the store floor, it often makes sense for a jewelry-needing customer to take some time to shop for complimentary items, rather than leaving and returning later. Overall, the STS service drives additional customers onto the BM sales floor. Thus, we expect sales associates will convert this increased store traffic into additional revenue by cross-selling additional products and services to STS customers.

Hypothesis-3 (H3). Introducing STS will increase BM channel sales.

We expect BM channel sales will also increase due to an increase in RTS incidents. Consumer returns are a service product offered by the retailer. Thus, a technology development such as STS will lower transaction costs of making returns as well, increasing demand for returns (Tyagi, 2004). Note that one of the main objectives of STS is to increase BM store traffic by bringing online customers into physical stores and thereby to sell additional products and warranty services to those customers. Interestingly, the same phenomenon also applies to consumer returns. If a return takes place at one of the physical stores, then the retailer acquires another opportunity to sell additional products and warranty services to those customers. With an increase in STS in-store return events, we expect store sales to increase. During a return, the operating procedures of the retailer that we study instruct sales associates to try to convert returns into exchanges or purchases of other items. Again, on the retailer's side, the store traffic that arises from returns of online purchases to physical stores reduces transaction costs of acquiring selling opportunities. From a customer's perspective, the marginal cost of shopping for alternate items is also nearly zero, given that they are already in the store. Thus, we expect that introducing STS will not only increase sales due to increased store traffic at the time of pick-up (i.e., H3), but also due to increased store traffic at the time of returns.

Hypothesis-4 (H4). Introducing STS will increase BM channel sales indirectly through an increase in RTS incidents.

4. Analyzing the impact of STS

While it is typical to organize empirical research in a sequence that first describes the data and defines all variables of interest, second introduces and defines all of the requisite models used for estimation, and third presents all of the estimation results, that is not the approach we take here. Our research addresses two different channels that involve models with different units of analysis and different control variables. Each analysis has multiple models with different dependent variables. Thus, discussing everything all at once would be a disservice to the reader. Instead, we separate our analyses for the online and BM channels to simplify the exposition. Hence, for each channel, we describe the associated data, models, and results separately. Within the analysis of the online channel, we separately discuss the impacts of STS on sales versus on returns. In this section, we first describe details of the retailer's STS service in §4.1, then report our analysis of the impact of STS with respect to the online channel in §4.2, report our analysis with respect to the BM channel in §4.3, and finally discuss the results and potential causal factors in §4.4.

4.1. Retailer channel details and analysis approach

Retailer Channel Details We collected data from a national jewelry retailer that implemented STS. The retailer offers a wide variety of brand-name jewelry, watches, accessories, and service plans to its customers. This corporation has more than 1000 BM stores in the U.S. and Canada. We note that the retailer we study operates BM stores for several of its different brands that include flagship brand stores, secondary brand stores, and outlet stores. Our analysis focuses on flagship brand stores in the U.S. During the period of our empirical analysis, the retailer operated more than 600 flagship store locations in the U.S.

In addition to BM stores, the retailer has operated an online channel for its flagship brand since the late 1990s. During the mid- and late-2000s, the retailer expanded its e-commerce business by launching website channels for several of its secondary brands and outlet brands. The retailer implemented STS into its online channel systems for its flagship brand in mid-2011.

STS Details Before STS was introduced, the retailer had offered its online customers several different fee-based shipping options. After STS was introduced, customers gained the new option to select between a fee-free STS option and the same fee-based shipping alternatives that had been previously offered. Now, if customers choose STS, their orders are shipped from a central warehouse to their preferred store location within three business days. We note that BM stores that receive STS delivery of purchases for customer pickup receive credit for those online sales. During the time period under study, to the best of our knowledge, no other shipping policies changed.

The first STS transaction took place on August 1, 2011. STS apparently was initially introduced in California prior to the national rollout, as evidenced by 535 STS transactions in August 2011, with nearly 90% of them in California. In September 2011, we observe a meaningful number of STS transactions (3629 STS transactions) indicative of a nationwide rollout of the STS service for all stores. Hence, we use September 2011 as our start date for the STS service. For robustness purposes, we also performed an analysis using August 2011 as the start date and the results are consistent with what we report here.

Data Set Details Our raw transaction data includes all purchase and return transactions for customers at all of the jewelry retailer's online and BM store channels, for all brands, from August 1, 2009 to July 31, 2013, which extends across four fiscal years. Customers are uniquely identified with a customer ID so purchase and return transactions can be tracked over time by customer. To simplify our empirical analysis, we focus our attention on the flagship brand, which constitutes nearly 70% of stores and 73% of the jeweler's sales in the U.S. We describe robustness analyses in the Appendix using data from non-flagship brands to further validate the findings.

With September 2011 as the STS nationwide rollout date for all stores, we use data one year prior to and one year after September 2011 to evaluate the impact of STS on customer actions and retailer operations (i.e., September 2010 to September 2012). The cleaned data set thus includes all purchase transactions from September 2010 to September 2012 and all associated returns transactions for those customers' purchases at both online and BM store channels. Consistent with Song et al. (2015), we expect customers need a warm-up (acclimation) period to be familiarized with the new STS service. Thus, we specify September 2011 as the warm-up period and do not use the data from this month in our analyses. We note that we also have customer returns transaction data for 10 months (i.e., nearly 300 days) after September 2012, much longer than the retailer's standard 100 day return policy. This additional data allows us to track all returns, meaning we have no salient data censoring problems arising from the structure of the data set.

Empirical Analysis Approach A simple approach to investigate the impact of STS would examine differences between the variables of interest before and after STS is introduced using indicator variables. However, this approach may not generate accurate results, since other factors also may have influenced the changes. For example, changing trends in the variables of interest (e.g. increasing online sales or decreasing store sales) or a contemporaneous shock to the economy may drive such changes. Moreover, ignoring endogeneity may lead to biased inferences (Ketoviki and Guide, 2015). To account for such challenges and to protect against problems associated with endogeneity, we employ a difference-in-difference (DID) approach to answer our research questions. Note that the DID methodology can address many of the endogeneity problems arising from comparisons between heterogeneous experimental units by forming treatment and control groups and comparing the change in the response variable accordingly (Bertrand et al., 2004).

DID is used to estimate the impact of policy interventions and requires two separate subpopulations: treatment group and control group (Donald and Lang, 2007). Forming treatment and control groups addresses the concern that unobserved factors may be driving any potential correlation between the dependent and independent variables, which is one of the major concerns for endogeneity. Furthermore, in DID, the change in the treatment group, which is subject to a policy intervention, is adjusted by the change in the control group, which is not affected by the intervention. With both groups, the DID approach accommodates and controls for trends in the data that may otherwise confound the analysis (Athey and Imbens, 2006). In the context of STS introduction, once the treatment and control groups are identified, one can apply DID and measure the impact of the STS treatment by comparing the differences between the treatment and control groups before and after STS is introduced to customers. DID is a preferred methodology to evaluate a policy change when the data set is structured as a quasi-experiment and where control and treatment groups can be clearly identified. DID methodology has been previously used in related operations management literature (e.g., Caro and Gallien, 2010; Gallino and Moreno, 2014; Song et al., 2015).

4.2. Analysis of the online channel

Our main objective is to understand how introduction of STS impacts sales and returns in the retailer's online channel. To do so, we must establish control and treatment groups for our DID analysis. A natural way to define these groups is by taking into account the distance between online customers and the retailer's closest stores. Basically, STS is only viable for a customer if a store is nearby, since customers will need to travel to a local store to pick up their STS purchases. Hence, the treatment group consists of geographic areas with stores that are in close proximity. Similarly, the control group corresponds to geographic areas that are not in close proximity to the jewelry retailer's stores, and therefore are not affected by STS.

A convenient way to establish these groups is to use a designated-market-area (DMA)[1] as the unit of analysis. This approach is consistent with the approach of Gallino and Moreno (2014), which investigates BOPS. There are 210 DMAs in the U.S. We choose 50 miles as the cut-off threshold for classifying DMAs as either treatment or control. While 50 miles is somewhat arbitrary, it is large enough to make the STS service inconvenient and unattractive to customers in some DMAs. Hence, a DMA is in the treatment group if the median distance between customers and their closest stores in the DMA is less than 50 miles, otherwise the DMA is in the control group. Using this approach, we have 185 DMAs in the treatment group and 25 DMAs in the control group. For robustness analyses, we investigate different distances (30, 35, 40, 45 or 60 miles) and observe no differences in the findings at these distances. We also create treatment and control groups using alternate approaches, and our findings remain consistent, as described in the Appendix.

4.2.1. Variables and summary statistics

Table 2 reports summary statistics (means and standard deviations) for key variables of interest for all DMAs both before and after the STS rollout. Table 3 reports correlations between variables in our models.

Dependent Variables To investigate the impact of STS introduction on online channel sales and returns, we use total dollar sales (ONLINE SALES) and total dollar returns (ONLINE RETURNS) for each month and by U.S. DMA. Each variable was constructed from the jewelry retailer's transactional data. As a test of model robustness, we also replicate our analysis that uses ONLINE SALES as a dependent variable with a model that uses total number of purchase transactions (NUMBER OF PURCHASES) as a dependent variable. Similarly, we also replicate our analysis that uses ONLINE RETURNS as a dependent variable with models that use total number of return transactions (NUMBER OF RETURNS) and, separately, return rate (RETURN RATE), as dependent variables. Using these alternate dependent variables, we find that these results remain consistent with our primary approach, as described in the Appendix.

National Economic Activity Controls To alleviate concerns arising from economic conditions during the analysis period, we include several economic activity controls. To represent aggregate U.S. economic activity, we include variables for quarterly percent change in gross domestic product (GDP), monthly percent change in consumer price index (CPI), and monthly percent change in industry-level retail sales (IND. RETAIL SALES). Each variable was calculated relative to the immediate prior month. Data for GDP, CPI, and IND. RETAIL SALES are from the Federal Reserve Bank of St. Louis database. We also include the percent change in quarterly U.S. e-commerce sales (ESALES) as an economic control variable. Quarterly e-commerce data were obtained

[1] A DMA is a region where the population can receive the same (or similar) television and radio station offerings, and may also include other types of media including newspapers and Internet content. They can coincide or overlap with one or more metropolitan areas, though rural regions with few significant population centers can also be designated as markets. DMAs are widely used in audience measurements, which are compiled in the U.S. by Nielsen Media Research both for television and radio (from Wikipedia).

Table 2
Summary statistics for the online channel by DMA.

	Treatment (affected DMAs)				Control (Unaffected DMAs)			
	pre-STS		post-STS		pre-STS		post-STS	
Time-Variant Variables	Mean	SD	Mean	SD	Mean	SD	Mean	SD
ONLINE SALES (000)	30.84	60.89	35.02	68.88	4.68	6.33	5.79	7.41
ONLINE RETURNS (000)	4.79	9.35	5.59	11.47	0.72	1.21	0.85	1.30
NUMBER OF PURCHASES	176.11	374.54	193.66	415.07	23.62	33.02	27.79	39.23
NUMBER OF RETURNS	19.15	39.46	20.94	46.09	2.48	3.21	2.78	3.80
PROMOTION	0.86	0.10	0.88	0.10	0.84	0.20	0.88	0.15
S.STOCK	41.74	4.38	45.51	1.92	41.74	4.38	45.51	1.92
B.STOCK	49.49	6.55	35.45	5.25	49.49	6.55	35.45	5.25
T.STOCK	65.05	9.72	63.88	7.05	65.05	9.72	63.88	7.05
IND. RETAIL SALES	0.64	0.44	0.38	0.69	0.64	0.44	0.38	0.69
CPI	0.31	0.13	0.16	0.21	0.31	0.13	0.16	0.21
GDP	1.33	1.78	2.43	1.48	1.33	1.78	2.43	1.48
ESALES	3.75	0.67	3.87	1.38	3.75	0.67	3.87	1.38
Time-Invariant Variables	Mean		SD		Mean		SD	
DMA UNEMPLOYMENT (%)	5.01		1.82		4.59		1.36	
DMA RETAIL SALES (000)	28,595.37		40,084.17		5,818.83		4,673.34	

Table 3
Correlations of variables for the online channel.

		1	2	3	4	5	6	7	8	9	10	11	12	13	14
1.	ONLINE SALES	1.00													
2.	ONLINE RETURNS	0.86	1.00												
3.	HOLIDAY	0.28	0.18	1.00											
4.	DMA UNEMPLOYMENT	−0.03	−0.03	$0.00^\dagger$	1.00										
5.	DMA RETAIL SALES	0.68	0.62	$-0.01^\dagger$	−0.06	1.00									
6.	SUMMER	−0.09	−0.04	−0.14	$0.00^\dagger$	$0.01^\dagger$	1.00								
7.	PROMOTION	0.21	0.12	0.36	$0.00^\dagger$	$0.01^\dagger$	−0.22	1.00							
8.	S.STOCK	0.05	0.04	$0.00^\dagger$	$0.00^\dagger$	$0.00^\dagger$	0.08	0.13	1.00						
9.	B.STOCK	0.06	0.05	0.22	$0.00^\dagger$	$-0.01^\dagger$	0.34	$0.02^\dagger$	−0.21	1.00					
10.	T.STOCK	$0.00^\dagger$	0.04	$-0.01^\dagger$	$-0.01^\dagger$	$0.00^\dagger$	0.09	0.03	0.43	0.13	1.00				
11.	IND. RETAIL SALES	$-0.01^\dagger$	$0.01^\dagger$	−0.07	$0.00^\dagger$	$0.00^\dagger$	−0.39	$-0.02^\dagger$	−0.21	0.37	−0.13	1.00			
12.	CPI	−0.04	$-0.02^\dagger$	−0.04	$0.00^\dagger$	$0.00^\dagger$	−0.42	−0.06	0.05	0.52	0.06	0.56	1.00		
13.	GDP	0.11	0.08	0.30	$0.00^\dagger$	$0.00^\dagger$	−0.09	0.26	−0.07	−0.22	0.29	−0.12	−0.40	1.00	
14.	ESALES	0.10	0.08	0.28	$0.00^\dagger$	$-0.01^\dagger$	−0.30	0.15	−0.18	0.29	0.10	0.39	0.04	0.41	1.00

Note: Except for cells with †, all coefficients are significant at $p < 0.05$ level.

from the United States Census Bureau. We also considered other potential economic control variables such as monthly nationwide jewelry sales, available from the United States Census Bureau. This variable was very highly correlated with month-based seasonality controls (i.e., HOLIDAY), thus we omit it from the regressions.

Regional Retail Activity Controls We include two DMA-specific control variables that may influence the dependent variables in our models. Total DMA retail sales (DMA RETAIL SALES) denotes the total amount of retail sales in each DMA and controls for relative sizes of DMAs. Unemployment rate (DMA UNEMPLOYMENT) indicates the fraction of the eligible population that is unemployed in each DMA and controls for economic differences between DMAs. These two controls were obtained from annual reports published by Nielsen Media Research.

The jeweler's transaction data also provides an opportunity to observe whether a transaction occurred with a discount offer or promotional event. These offers include marketing coupons, store coupons, military discounts, gift certificates, and promotional events, among others. Using this information, we construct the variable PROMOTION for each DMA d during month t to represent the fraction of purchases that take place in the DMA with a discount offer. The PROMOTION variable helps isolate the impact of promotional events and discounts on our dependent variables. We also considered other potential regional control variables such as total adult population, average income, and buying power index, but these variables are highly correlated with DMA UNEMPLOYMENT and DMA RETAIL SALES. Hence, we do not include them in our models.

Competitor Controls To control for business activities of direct competition, we include monthly stock prices for major competitors such as Signet (S.STOCK), Blue Nile (B.STOCK), and Tiffany & Co. (T.STOCK). The stock price information for the three major competitors is taken from Yahoo Finance.

We also considered including the number of jewelry stores in each DMA. This variable was calculated by using data on more than 5000 jewelry stores across the U.S. and allocated to each DMA across years. The locations of jewelry stores across the U.S. during the period of analysis were acquired from AggData, which sells data about current and historical retail store locations. However, we found the number of jewelry stores in each DMA is highly correlated with DMA RETAIL SALES. Hence, we do not include this variable in our online channel models.

Seasonality Controls We employ indicator variables as seasonal controls. These variables include whether a transaction occurred during the holiday season (HOLIDAY) or during the early summer period (SUMMER). HOLIDAY spans all of December, which in addition to covering several religious holidays, is also the most popular month for couples to get engaged to be married (TheKnot, 2013). SUMMER covers the months of June and July, which corresponds to months that are

Table 4
Impact of STS on online channel sales.

Variables	Online sales		High value sales		Low value sales	
GROUP	0.45***	(0.084)	0.41***	(0.079)	0.47***	(0.083)
POLICY	0.24***	(0.036)	0.09	(0.053)	0.17**	(0.054)
GROUP*POLICY	−0.14***	(0.038)	−0.15***	(0.039)	−0.02	(0.042)
HOLIDAY	1.14***	(0.023)	1.05***	(0.028)	1.26***	(0.027)
SUMMER	−0.22***	(0.017)	−0.26***	(0.025)	−0.39***	(0.024)
PROMOTION	1.06***	(0.059)	0.90***	(0.057)	0.66***	(0.046)
DMA RETAIL SALES	0.99***	(0.024)	0.98***	(0.023)	1.00***	(0.022)
DMA UNEMPLOYMENT	0.02	(0.014)	0.02	(0.013)	0.01	(0.013)
S.STOCK	0.03***	(0.003)	0.03***	(0.003)	0.02***	(0.003)
B.STOCK	0.01***	(0.002)	0.01**	(0.002)	0.02***	(0.002)
T.STOCK	−0.01***	(0.001)	−0.01***	(0.001)	−0.01***	(0.001)
IND. RETAIL SALES	0.06***	(0.015)	0.05***	(0.016)	0.12***	(0.015)
CPI	−0.65***	(0.051)	−0.61***	(0.055)	−0.86***	(0.054)
GDP	0.00	(0.006)	0.00	(0.006)	0.02***	(0.005)
ESALES	0.01	(0.009)	0.03	(0.009)	−0.05***	(0.009)
N	5,024		5,009		4,966	
R^2	0.59		0.56		0.63	
HYPOTHESIS	H1A		H1B		H1B	
SUPPORT	No		No		No	

Note: Standard errors are reported in parentheses.
**p* < 0.05, ***p* < 0.01, ****p* < 0.001.

among the most popular in which to get married (TheKnot, 2013). We also estimated models that included a full set of monthly indicator variables, wherein model estimates were consistent with the results presented here and with roughly the same R^2.

We note that none of the correlations between independent variables have levels close to or higher than 0.80, which minimizes concerns about multicollinearity (Song et al., 2015). We also check for multicollinearity by computing variance inflation factors (VIF). Results show that the largest VIF is 2.16 and the mean VIF is 1.54, thus compared to the suggested cut-off level of ten (Wooldridge, 2012), we do not expect multicollinearity to be an issue. Even so, blindly using a pure cut-off level without further consideration of the sample size and context of the data analysis is not recommended (Ketoviki and Guide, 2015). Considering all elements, however, we feel comfortable dismissing multicollinearity as a concern.

Our explanatory variables include indicator variables that denote whether DMA d is within a store's area of influence ($GROUP_d$) and if the observation belongs to the time period after the STS policy was introduced ($POLICY_t$). These indicator variables are given by:

$$GROUP_d = \begin{cases} 1 & \text{if median customer distance to closest BM stores in } DMA_d < 50 \text{ miles} \\ 0 & \text{otherwise} \end{cases}$$

$$POLICY_t = \begin{cases} 1 & \text{if } t > \text{september } 2011 \\ 0 & \text{otherwise} \end{cases}$$

We capture the introduction of STS with a binary interaction term, $GROUP_d{*}POLICY_t$, which is equal to 1 for the treatment DMAs after the rollout of STS, and is 0 otherwise. The estimation of a coefficient for this variable measures the impact of STS on the dependent variable in our models. Hence, $GROUP_d{*}POLICY_t$, is the key variable in our models used to examine hypotheses.

One potential concern is that store openings and closings might affect the median distance of customers to the closest stores within a DMA. First, for the study's time period, we verified that there are no store openings or store closings in the control group DMAs. Next, for the treatment group DMAs, we calculate the median distance of customers to all of the stores that are open, by month and by DMA. We find there is no treatment DMA in which changes to the median distance changes a DMA's classification between control group and treatment group. In effect, neither store openings nor store closings change the designation of a DMA to the treatment versus control groups in the data set, thus our findings are not affected by this concern.

4.2.2. Models, estimation, and results for online sales

To examine H1A, we use the following model specification:

$$\log(\text{ONLINE SALES}_{dt}) = \mu_d + \alpha_1 * \text{GROUP}_d + \alpha_2 * \text{POLICY}_t + \alpha_3 * \text{GROUP}_d * \text{POLICY}_t + \alpha_m * \text{CONTROLS}_{mdt} + \varepsilon_{dt} \quad (1)$$

Where the dependent variable denotes the log of total online product sales within DMA d during month t. $CONTROLS_{mdt}$ denotes the vector of control variables including HOLIDAY, SUMMER, PROMOTION, DMA RETAIL SALES, DMA UNEMPLOYMENT, S. STOCK, B. STOCK, T. STOCK, IND. RETAIL SALES, CPI, GDP, and ESALES while α_m is a vector of coefficients that correspond to control variables. Note that index m represents the size of the vector for control variables.

Although our DMA-level controls are time-invariant, we expect that DMA RETAIL SALES and DMA UNEMPLOYMENT can explain cross-DMA variation in the dependent variables. We also conduct a Hausman specification test, which rules against the fixed-effects (FE) model in favor of the consistent and more efficient random-effects (RE) model ($\chi^2 = 0.63$, $p > 0.05$). Hence, we use RE models for estimation, which have been shown to be asymptotically efficient for our setting (Donald and Lang, 2007).

To investigate the impact of STS on sales of high-value and low-value online products, we use $98.69 as a threshold to create the two groups. This value is the median for online sales and is also close to what the jewelry retailer considers to be the threshold for high-value items ($100), as indicated by the retailer's shipping policy. For H1B, we apply equation (1) separately to both the high-value sales group and the low-value sales group.

Our results are reported in Table 4. Each column (separated by vertical lines) in Table 4 examines the hypotheses listed in the second to last row of the table. The first row of each column identifies the dependent variable for the corresponding regression model. For each column, we report the results for the variable of interest (*GROUP**-*POLICY*) in the fourth row. Note that the number of observations, *N*, is slightly different for each column. This minor difference arises because there may be no high-value/low value product purchases for a DMA during a specific month.

The results for H1A are reported in the second column of Table 4.

After the STS introduction, there is a negative and significant effect ($\alpha_3 = -0.14$, $p < 0.001$) on the retailer's online sales in the affected STS treatment group DMAs ($GROUP_d{*}POLICY_t == 1$) relative to those DMAs that are not within the influence area of a physical store ($GROUP_d{*}POLICY_t == 0$). This result suggests introducing STS reduced the retailer's online channel sales. Hence, H1A is not supported. While DID methodology is intended to account for potential confounding due to endogeneity, a possible concern is the potential for different trends for the control and treatment groups before STS is implemented. To address this issue, we report a pre-intervention trend analysis in the Appendix for each model, supporting our findings.

The H1B results are reported in the third and fourth columns of Table 4. We find the effect of introducing STS on the retailer's online sales of high-value products in affected treatment group DMAs is negative and significant ($\alpha_3 = -0.15$, $p < 0.001$) relative to DMAs that are not affected. Our results also show that STS introduction does not have a significant impact ($\alpha_3 = -0.02$, $p > 0.05$) on the retailer's online sales of low-value products in affected DMAs relative to unaffected DMAs. Hence, the findings do not support H1B.

4.2.3. Models, estimation, and results for online returns

Here, we analyze the impact of STS on product returns. We first introduce our model to examine the hypothesis related to online returns (H2B) and then we introduce our model to evaluate online product returns that are returned to stores (H2A). We present in this order since the model for H2A adds a complication not present in the other model.

For H2B, which describes the relationship between STS implementation and online product returns, we use the following model specification:

$$\log(\mathrm{ONLINE\ RETURNS}_{dt}) = \mu_d + \beta_1 * \mathrm{GROUP}_d + \beta_2 * \mathrm{POLICY}_t + \beta_3 * \mathrm{GROUP}_d * \mathrm{POLICY}_t + \beta_4 * \log(\mathrm{ONLINE\ SALES}_{dt}) + \beta_m * \mathrm{CONTROLS}_{mdt} + \varepsilon_{dt} \quad (2)$$

Where our dependent variable represents the log of online product returns at DMA d during month t. We also include log ($ONLINE\ SALES_{dt}$) as a control variable since sales influence the amount of returns.

We next analyze the impact of STS on online purchases that are returned to stores (H2A). We find over 60,000 online transactions in our data that have been returned to a BM store. Because the DID analysis requires a large number of observations, we use state-month as our unit of analysis in this model, rather than DMA-month. Since we have 210 DMAs, DID methodology will have, on average, 12 purchase transactions per month for each DMA ($60{,}000/(210 * 24) = 11.9$). With a state-level analysis, however, our model will have on average 50 purchase transactions per month for each state ($60{,}000/(50 * 24) = 50$) and will provide enough statistical power to detect differences if they exist. Therefore, we use a state level analysis. Table 5 reports cross returns summary statistics (means and standard deviations) for key variables of interest for all states both before and after the STS rollout.

A challenge with state-level analysis is that state populations are more densely located in large metropolitan areas, precisely where most stores also are located. As a result, our ability to separate out populations that are unaffected by introduction of STS is constrained. If we use 50 miles as our threshold distance for classification of states, as we did for DMAs, we would have only one state in the control group. Instead, we choose 35 miles. Note that choosing a shorter distance threshold provides a more stringent test of hypotheses since the influence of STS diminishes with distance. Hence the shorter the distance, the more one should expect the control group to behave like the treatment group and consequently the more difficult it will be to observe statistically significant differences. With 35 miles, there are 4 states in the control group. This does not present a problem, due to the many individual customer observations within each state. We repeated the same analysis with threshold distances of 30, 25, 20, and 15 miles, all of which increase the number of states in the control group. With 15 miles, there exist 15 states in the control group and 35 states in the treatment group. Our findings remain consistent with those we report here.

For H2A, which describes the relationship between STS introduction and RTS incidents, we use the following model specification:

$$\log(\mathrm{CROSS\ RETURNS}_{kt}) = \mu_k + \beta_1 * \mathrm{GROUP}_k + \beta_2 * \mathrm{POLICY}_t + \beta_3 * \mathrm{GROUP}_k * \mathrm{POLICY}_t + \beta_4 * \log(\mathrm{ONLINE\ SALES}_{kt}) + \beta_m * \mathrm{CONTROLS}_{mkt} + \varepsilon_{dt} \quad (3)$$

Where our dependent variable represents the log of online product sales returned to a store in state k during month t. As in equation (2), we include log ($ONLINE\ SALES_{kt}$) as a control, allowing us to evaluate the effect of STS on consumer returns at DMAs that do not arise from a change in online product sales.

The results for H2A using the estimation of equation (3) are reported in the second column of Table 6. We observe the effect of STS introduction on the RTS behavior in the affected states is positive and significant ($\beta_3 = 0.52$, $p < 0.001$) compared to unaffected states. This finding supports H2A.

The results for H2B, using equation (2), are reported in the third column of Table 6. We observe that the effect of STS on the retailer's customer returns in the affected DMAs is not significant ($\beta_3 = 0.11$, $p > 0.05$) compared to DMAs that are not affected. Hence, we do not find support for H2B.

4.3. Analysis of the brick-and-mortar channel

Next, we investigate the impact of STS introduction on BM stores. We again employ a DID approach, but here the treatment group consists of U.S. stores while the control group consists of Canadian stores. This DID classification arises because STS was implemented in the U.S. but not immediately in Canada (only later). A concern might be raised that any observed differences might be explained by cultural or economic differences between the two countries, rather than the impact of STS. To alleviate this concern, we also repeated our analysis using a different treatment group from Canada. This alternate treatment group consists of stores that the retailer operates under a different brand name and for which STS was implemented a year after it was introduced in the U.S. The results from that analysis, reported in the Appendix, corroborate

Table 5
Summary statistics for cross returns analysis.

	Treatment (Affected States)				Control (Unaffected States)			
	pre-STS		post-STS		pre-STS		post-STS	
Time-Variant Variables	Mean	SD	Mean	SD	Mean	SD	Mean	SD
CROSS RETURNS (000)	6.50	8.64	8.95	12.17	0.87	0.85	0.99	1.42
NUMBER OF CROSS RETURNS	24.87	32.68	31.73	44.61	2.76	2.20	2.84	2.62
STATE UNEMPLOYMENT (%)	8.48	1.92	7.63	1.73	6.68	1.23	5.96	1.26
PROMOTION	0.86	0.19	0.89	0.14	0.85	0.31	0.90	0.27

Table 6
Impact of STS on online channel returns.

Variables	Cross returns		Online returns	
GROUP	0.09	(0.144)	−0.25***	(0.063)
POLICY	−0.24	(0.184)	0.03	(0.093)
GROUP*POLICY	0.52***	(0.154)	0.11	(0.073)
ONLINE SALES	1.11***	(0.032)	1.07***	(0.024)
HOLIDAY	−0.21*	(0.086)	−0.22***	(0.051)
SUMMER	0.09	(0.073)	0.21***	(0.041)
PROMOTION	0.40***	(0.113)	−0.45***	(0.117)
DMA RETAIL SALES	–	–	0.02	(0.026)
DMA UNEMPLOYMENT	0.06***	(0.016)	−0.00	(0.008)
S.STOCK	−0.01	(0.010)	−0.02**	(0.006)
B.STOCK	−0.01	(0.005)	0.00	(0.003)
T.STOCK	0.00	(0.003)	0.01***	(0.002)
IND. RETAIL SALES	−0.05	(0.045)	0.05	(0.025)
CPI	0.43**	(0.160)	0.25**	(0.090)
GDP	−0.02	(0.017)	−0.00	(0.010)
ESALES	−0.01	(0.027)	−0.01	(0.015)
N	1,160		4,667	
R^2	0.64		0.64	
HYPOTHESIS	H2A		H2B	
SUPPORT	Yes		No	

Note: Standard errors are reported in parentheses.
$*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

the findings we report here.

We note that there are 11 store openings and 50 store closings throughout the period of analysis that could potentially distort our results. To alleviate this concern, we only include stores that are open both pre-STS and post-STS. Doing so leaves 602 stores from the U.S. and 51 stores from Canada. However, as we will discuss, we do account for the potential impact of a store closure on our estimates.

4.3.1. Variables and summary statistics

Dependent Variables To investigate the impact of STS introduction on BM sales and returns, as shown above, we again use total dollar sales and total dollar returns for each month and store in the U.S. and Canada. To test model robustness with respect to our choice of dependent variables, we also replicate our analysis using total number of purchase transactions, total number of return transactions, and return rate for each month and store as alternate dependent variables. Our results using these alternative dependent variables are consistent with our primary analysis as presented in the Appendix. Table 7 reports the summary statistics (mean and standard deviation) for these variables of interest.

Control Variables Previously Described As in the online channel analysis, we also employ HOLIDAY, SUMMER, PROMOTION, T. STOCK, B. STOCK, S. STOCK, IND. RETAIL SALES, GDP, and CPI as control variables. In addition to these variables, we include several new controls.

National Economic Activity Controls We include an economic control variable, UNEMPLOYMENT, which indicates the monthly unemployment rate for U.S. states and Canada provinces. We obtained unemployment information from the U.S. Bureau of Labor Statistics and from Statistics Canada.

Store Controls We control for store characteristics using five variables. MALL GRADE is a categorical variable used by the retailer to classify the location of stores based on a mall's total sales area, number of stores, customer traffic, and annual sales revenue. MALL GRADE has ten levels, which include community centers, regional centers, power centers, lifestyle centers, metro centers, and village locations, reported as a set of ordinal values such as A+, A, B, C, and F. We obtained MALL GRADE information both from the retailer that we study and from the Directory of Major Malls. Next, inventory volume group (VOLUME GROUP) is an ordinal variable that the retailer uses to classify its stores based on the annual sales revenue (e.g. $1-$1.5 million corresponds to B). VOLUME GROUP has five levels, which include ordinal values of A+, A, B+, B, and C. Store size (STORE SIZE) denotes the physical square footage of each store. Number of cases (CASE COUNT) is an inventory display device used by stores. Inventory turnover (INVENTORY TURNS) denotes the number of times inventory is sold and replaced in a year for each store.

Competitor Controls We employ a count of the number of competitor jewelry stores (stores) for each store location (of the retailer we study) across years to control for competition. The number of local competitor jewelry stores for each store location across years is calculated using data from AggData on around 6000 jewelry stores across the U.S. and Canada, and allocated to our retailer's stores by using their ZIP codes. We also include the indicator CLOSURE to control for the impact of a store closure on the nearest existing store location. Overall, 50 store

Table 7
Summary statistics for the BM channel by store.

	Treatment (U.S. stores)				Control (Canada stores)			
	pre-STS		post-STS		pre-STS		post-STS	
Time-Variant Variables	Mean	SD	Mean	SD	Mean	SD	Mean	SD
STORE SALES (000)	113.58	88.04	127.07	98.52	102.75	72.96	104.32	74.75
STORE RETURNS (000)	20.60	19.09	22.67	20.42	16.62	13.58	17.46	14.61
NUMBER OF PURCHASES	277.59	231.95	310.28	266.83	368.79	251.32	389.46	299.35
NUMBER OF RETURNS	34.36	26.82	38.35	31.72	38.57	27.31	39.55	30.38
PROMOTION	0.91	0.05	0.84	0.08	0.69	0.09	0.68	0.10
UNEMPLOYMENT	8.97	1.95	8.08	1.73	7.72	1.25	7.31	1.39
S.STOCK	41.74	4.38	45.51	1.92	41.74	4.38	45.51	1.92
B.STOCK	49.49	6.55	35.45	5.25	49.49	6.55	35.45	5.25
T.STOCK	65.05	9.72	63.88	7.05	65.05	9.72	63.88	7.05
IND. RETAIL SALES	0.64	0.44	0.38	0.69	0.35	0.66	0.19	0.54
CPI	0.31	0.13	0.16	0.21	0.25	0.27	0.10	0.27
GDP	1.33	1.78	2.43	1.48	1.70	0.49	0.62	0.56
STORES	1.58	1.46	2.54	2.07	0.69	0.75	0.88	0.83
CLOSURE	0.02	0.14	0.05	0.21	0.02	0.13	0.05	0.22

Time-Invariant Variables	Mean	SD	Mean	SD
STORE SIZE (sqft)	1,682.53	418.19	1,528.28	293.92
INVENTORY TURNS	0.90	0.15	0.83	0.16
CASE COUNT	33.02	4.96	30.14	4.38

Table 8
Correlations of variables for the BM channel.

		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
1.	STORE SALES	1.00																
2.	STORE RETURNS	0.84	1.00															
3.	HOLIDAY	0.49	0.34	1.00														
4.	CASE COUNT	0.14	0.14	0.00	1.00													
5.	STORE SIZE	0.23	0.22	0.00	0.25	1.00												
6.	INVENTORY TURNS	0.41	0.32	0.00	0.02	0.11	1.00											
7.	UNEMPLOYMENT	−0.01†	0.06	0.03	0.09	0.10	−0.10	1.00										
8.	SUMMER	−0.13	−0.12	−0.13	0.00†	0.00†	0.00†	−0.04	1.00									
9.	PROMOTION	0.13	0.14	0.16	0.07	0.06	0.06	0.21	−0.14	1.00								
10.	S.STOCK	0.09	0.05	0.00†	0.00†	0.00†	0.00†	−0.16	0.09	−0.19	1.00							
11.	B.STOCK	0.08	0.06	0.21	0.00†	0.00†	0.00†	0.20	−0.34	0.37	−0.21	1.00						
12.	T.STOCK	0.00†	0.00†	−0.01†	0.00†	0.00†	0.00†	0.00†	0.11	0.08	0.44	0.13	1.00					
13.	IND. RETAIL SALES	−0.04	0.00†	0.06	0.02	0.01†	0.01†	0.09	−0.37	0.22	−0.22	0.36	−0.12	1.00				
14.	CPI	−0.07	−0.04	−0.04	0.01†	0.01†	0.01†	0.08	−0.43	0.23	0.03	0.50	0.06	0.49	1.00			
15.	GDP	0.18	0.15	0.29	0.02	0.01†	0.01†	0.01†	−0.09	0.09	−0.09	−0.18	0.27	−0.08	−0.34	1.00		
16.	STORES	0.17	0.16	−0.01†	0.07	−0.01†	0.17	−0.05	0.01†	0.12	0.02	−0.01†	0.01†	0.02	0.01†	0.03	1.00	
17.	CLOSURE	0.01†	0.00†	−0.02	0.06	0.05	0.04	−0.02	0.03	−0.03	0.07	−0.07	0.03	−0.04	−0.03	0.01†	0.01†	1.00

Note: Except for cells with †, all coefficients are significant at $p < 0.05$ level.

closings occur during the analysis period. Once a store is closed, we find the nearest existing store location and assign a value of one to CLOSURE from that point until the end of the analysis period, but only if the nearest existing store location is within 50 miles of the closed store. Otherwise, we assume that the store closure will not impact customer behavior in that region.

Table 8 reports correlations between variables used in this analysis. As for the online channel, we check for multicollinearity and find that it is not a concern for our analysis of the BM store channel.

4.3.2. Models, estimation, and results for BM sales

To examine H3 and evaluate the impact of STS on BM store sales, we use the following model setup:

$$\log(\text{STORE SALES}_{jt}) = \mu_j + \delta_1 * \text{GROUP}_j + \delta_2 * \text{POLICY}_t + \delta_3 * \text{GROUP}_j * \text{POLICY}_t + \delta_s * \text{CONTROLS}_{sjt} + \varepsilon_{dt} \quad (4)$$

where our dependent variable represents the log of total sales at store j during month t. The explanatory variables include indicator variables that denote if store j is located in the U.S. ($GROUP_j$), if the observation belongs to the time period after the STS implementation ($POLICY_t$), and an interaction term between these indicator variables. We capture the adoption of STS with a binary interaction term ($GROUP_j * POLICY_t$), which is equal to 1 for the U.S. stores after the introduction of STS and 0 otherwise. $CONTROLS_{sjt}$ denotes the vector of control variables, while δ_s is a vector of coefficients that correspond to control variables. Index s represents the size of the vector for control variables.

Although some store level control variables are time-invariant, these variables may explain cross-store variation in the dependent variables. Note a Hausman test again rejects the fixed-effects model in favor of the random-effects model ($\chi^2 = 9.91$, $p > 0.05$). Hence, we again use random effects models for estimation.

The results for H3, using equation (4), are reported in the second column of Table 9. We find that the impact of STS on the retailer's post-STS sales in the U.S. stores ($GROUP_j * POLICY_t == 1$) is positive and significant ($\delta_3 = 0.15$, $p < 0.001$) relative to Canadian stores. This finding shows that implementing STS increases store sales. Hence, H3 is supported.

We thus proceed to investigate BM store sales generated after RTS incidents. Given the results for H2A, which shows RTS incidents increase, to demonstrate support for H4, all we must show is that RTS incidents lead to new purchases in stores. Looking into our transactional data, we find about 17,000 customer instances in which there are one or more store purchase transactions on the same day as when an RTS incident occurs for that same customer. This finding shows 28% of all RTS incidents are converted into new sales, corresponding to $7.17 million in additional sales revenue. Hence, the finding provides support for H4.

Table 9
Impact of STS on BM channel.

Variables	Store Sales	
GROUP	−0.18***	(0.032)
POLICY	−0.18***	(0.018)
GROUP*POLICY	0.15***	(0.017)
HOLIDAY	1.02***	(0.009)
SUMMER	−0.21***	(0.008)
PROMOTION	0.91***	(0.039)
MALL GRADE A+	−0.00	(0.036)
MALL GRADE B	−0.05*	(0.023)
MALL GRADE C	−0.10***	(0.023)
MALL GRADE F	−0.18***	(0.030)
MALL GRADE-COM	−0.22*	(0.096)
MALL GRADE-LIF	−0.15**	(0.046)
MALL GRADE-MET	0.12	(0.109)
MALL GRADE-POW	−0.16	(0.131)
MALL GRADE-VIL	−0.10	(0.096)
VOLUME GROUP A+	0.24***	(0.049)
VOLUME GROUP B	−0.64***	(0.028)
VOLUME GROUP B+	−0.34***	(0.029)
VOLUME GROUP C	−0.98***	(0.030)
STORE SIZE	0.00*	(0.000)
INVENTORY TURNS	0.54***	(0.054)
UNEMPLOYMENT	0.00*	(0.003)
CASE COUNT	0.00**	(0.002)
S.STOCK	0.03***	(0.001)
B.STOCK	−0.00***	(0.000)
T.STOCK	−0.00***	(0.000)
IND. RETAIL SALES	0.02***	(0.005)
CPI	−0.45***	(0.016)
GDP	0.00**	(0.002)
STORES	0.00	(0.004)
CLOSURE	0.04	(0.020)
N	15,672	
R^2	0.64	
HYPOTHESIS	H3	
SUPPORT	YES	

Note: Standard errors are reported in parenthe-ses.
$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

4.4. Discussion of results

It is surprising to find that the first two hypotheses (H1A and H1B) are not supported. In contrast to the *a priori* expectation that STS introduction would increase online sales, and even more so would increase the amount of high-value online sales, the empirical results suggest the opposite phenomena occurred. Clearly, there is something unexplained and unaccounted for going on in the online channel or between the online and BM channels. We examine this quandary in the next section.

With respect to H2A, we do observe empirical support that STS introduction increases return-to-store events. Nevertheless, we find it interesting that RTS incidents increase when actually online sales decrease. As a possible explanation, we believe this result may arise because of a perceived increase in return policy leniency. STS makes returning online purchases easier for customers. Alternately, the result may indicate STS in and of itself is simply informing more online customers about nearby stores. Interestingly, for H2B we do not see a net increase of online shopping returns. Again, we find this a bit surprising given that RTS incidents significantly increase. We envision two possible explanations why online returns are not changing after the STS rollout. We surmise either the observed increase in RTS incidents is not measurable, with respect to the totality of returns, since RTS incidents represent a small portion of overall returns, or other types of online returns are decreasing.

Next, for the hypotheses regarding the impact of the introduction of STS on the BM store channel, with H3 we observe empirical support that the introduction of STS by the jewelry retailer led to higher relative BM store sales. When we further looked into the detailed customer transaction data about whether STS customer in-store returns were time-connected to same-day in-store BM purchases made by the same customer, we observed strong support for H4 via a sizable number of return-to-store events that contributed substantial additional sales revenues.

5. Extended analysis

So far our research has demonstrated a link between the STS rollout and an increase in BM channel sales, but the precise explanation for this finding remains elusive. With one channel demonstrating an increase in sales and another channel demonstrating a decrease in sales, there appears to be a channel shift in customer purchase activity after the introduction of STS. This finding is unexpected. The increase in BM sales should be driven by an increase in store traffic that arises from STS customer pickups and additional returns generated by a marginal increase in online purchases. Yet, that cannot be the explanation here since online sales actually decrease during the year after the STS rollout. Moreover, our hypothesis (H1B) that the increase in online sales should be greater for high-value items than for low-value items is not supported. After the rollout of STS, the *decrease* in online sales can mainly be attributed to high-value items. To clarify our understanding of what is happening both within and between channels, we conduct an extended analysis.

The simple fact that our results do not support our primary hypotheses raises a number of fundamental questions we intend to answer in this section. STS is designed to increase online sales. Through that increase, store sales should increase as well. Yet, that is not what we observe. Online sales decrease, although store sales do increase. There must be mechanisms at work here that are not fully explained by the reasoning behind our hypotheses. Does the increase in BM channel sales and the decrease in online sales simply represent a channel shift in buyer behavior? When considering the channels in aggregate, does the retailer observe a net increase or decrease in sales due to STS? What differentiates the customers that end up using STS from those that continue to use home delivery and those that buy in stores? Our analysis here is directed towards answering such questions. First, we dig deeper into the BM channel to better explain the increase in BM store sales that we observe after the STS rollout. Then, we perform an aggregate analysis to determine the overall effect of STS on sales. Finally, we conduct a comprehensive cross-channel analysis to gain a better understanding of how customers differentiate between channels.

5.1. Digging deeper into the BM channel

To begin, we want to clarify our understanding of the change that occurs in the BM channel due to the STS introduction. We know, statistically, that online sales of high-value items decrease and low-value items remain unchanged. Can the increase in BM sales be attributed to the decline in the online sale of high-value items? Similarly, we would like to know how BM returns are affected.

To proceed, we again employ DID to take a closer look at both sales and returns for the BM channel before and after the STS rollout, differentiating between low-value and high-value items. To be consistent with §4.2, we first use \$98.69 as a threshold to create the two groups of items. Subsequently, we also use \$150 since the median of BM channel sales is near \$150 (\$149.99). We note the findings are the same for the two approaches. Our dependent variables for the analysis of BM sales are the log of BM channel sales of high-value and low-value products at store *j* during month *t*. We again use the model specification in equation (4) and estimate it separately for both the high-value sales group and the low-value sales group.

Similarly, we also evaluate returns of high-value and low-value items at the BM channel. To do so, we use the following model. Here we include $STORE\ SALES_{jt}$ as a control variable. We report results in Table 10.

$$\log(\mathrm{RETURNS_{jt}}) = \mu_j + \delta_1 * \mathrm{GROUP_d} + \delta_2 * \mathrm{POLICY_t} + \delta_3 * \mathrm{GROUP_d} * \mathrm{POLICY_t} + \delta_4 * \log(\mathrm{STORE\ SALES_{jt}}) + \beta_s * \mathrm{CONTROLS_{sjt}} + \varepsilon_{dt} \quad (5)$$

Store-Level Analysis of Channel Shift on Sales We find the effect of introducing STS on the retailer's BM channel sales of high-value products in the U.S. stores is positive and significant ($\delta_3 = 0.21, p < 0.001$) relative to Canadian stores that are not affected by STS. We also show that STS does not have any statistically significant effect ($\delta_3 = -0.00$, $p > 0.05$) on the retailer's BM channel sales of low-value products in the U.S. stores relative to Canadian stores. These results, in light of what we observe for the online channel, provide evidence that customers do move from the online channel to the BM channel to purchase high-value products, while the retailer's customers have continued using the online channel to purchase low-value products. It is interesting to note that the channel shift for high-value items occurs even though in-store product availability is not provided to online customers and that many of the products offered online are not available in stores.

Customer-Level Analysis of Channel Shift on Purchases To further validate our findings with respect to channel switching, we conducted several more granular customer-level analyses. First, we tabulate the relative channel preference of each customer pre-STS by calculating the ratio of their count of BM purchases to their total number of purchases. Second, by following the same steps, we tabulate the channel preference of each customer post-STS. Next, we match the two customer sets and focus only on customers who made purchases both pre-STS and post-STS. The number of customers is around 226,000. Comparing their pre-STS and post-STS actions, we find about 209,000 (92.2%) customers have a higher BM channel preference after STS is introduced. We then restrict our analysis to customers who have more than four purchase transactions during the analysis period. This subset gives us around 72,000 customers. Again, comparing the pre-STS and post-STS actions, we find around 65,000 (90.0%) customers have a higher BM channel preference after STS is implemented. Finally, we focus on customers that purely used the online channel pre-STS, which is around 6500 customers. Observing their channel preference post-STS, we find that their use of the BM channel increases from 0% to 26.7%. Hence, we believe the results of this customer-level analysis with respect to observed channel use of customers pre-STS versus post-STS provide additional empirical evidence to support the customer channel switching that we observe via the DID methodology.

Store-Level Analysis of Channel Shift on Returns We also find that the effect of STS on the retailer's BM channel returns of high-value

Table 10
Impact of STS on BM channel sales and returns of high- and low-value items.

Variables	Sales				Returns			
	High		Low		High		Low	
GROUP	−0.23***	(0.033)	−0.85***	(0.042)	0.14***	(0.035)	0.09*	(0.037)
POLICY	−0.27***	(0.019)	−0.13***	(0.030)	0.19***	(0.029)	0.00	(0.043)
GROUP*POLICY	0.21***	(0.017)	−0.00	(0.027)	−0.10***	(0.027)	0.04	(0.039)
HOLIDAY	0.97***	(0.009)	1.52***	(0.015)	−0.41***	(0.018)	0.10***	(0.027)
SUMMER	−0.18***	(0.008)	−0.64***	(0.013)	0.04***	(0.013)	0.01	(0.021)
PROMOTION	1.39***	(0.052)	−0.08**	(0.028)	−0.42***	(0.079)	−0.14**	(0.040)
MALL GR. A+	−0.00	(0.036)	0.01	(0.048)	−0.01	(0.033)	−0.02	(0.035)
MALL GR. B	−0.05*	(0.023)	0.00	(0.030)	−0.02	(0.021)	−0.01	(0.022)
MALL GR. C	−0.11***	(0.024)	0.05	(0.032)	−0.05*	(0.022)	−0.05*	(0.024)
MALL GR. F	−0.20***	(0.031)	0.09*	(0.041)	−0.08**	(0.029)	−0.07*	(0.031)
MALL GR.-COM	−0.23*	(0.097)	0.20	(0.129)	−0.13	(0.089)	−0.21*	(0.096)
MALL GR.-LIF	−0.14**	(0.047)	−0.06	(0.062)	−0.03	(0.043)	−0.03	(0.046)
MALL GR.-MET	0.13	(0.110)	0.29*	(0.147)	−0.05	(0.101)	0.07	(0.106)
MALL GR.-POW	−0.15	(0.134)	0.01	(0.177)	−0.19	(0.122)	−0.17	(0.135)
MALL GR.-VIL	−0.13	(0.098)	0.34**	(0.129)	−0.04	(0.089)	−0.15	(0.094)
VOL. GR. A+	0.25***	(0.050)	0.23***	(0.066)	0.00	(0.045)	0.08	(0.047)
VOL. GR. B	−0.65***	(0.029)	−0.36***	(0.038)	0.08**	(0.027)	−0.02	(0.028)
VOL. GR. B+	−0.34***	(0.030)	−0.18***	(0.040)	0.02	(0.028)	−0.01	(0.029)
VOL. GR. C	−0.98***	(0.031)	−0.56***	(0.041)	0.15***	(0.031)	−0.00	(0.031)
STORE SIZE	0.00	(0.000)	0.00**	(0.000)	0.00	(0.000)	0.00***	(0.000)
INVEN. TURNS	0.54***	(0.055)	0.45***	(0.073)	−0.33***	(0.051)	0.02	(0.054)
UNEMP. RATE	−0.00	(0.004)	0.00	(0.005)	0.03***	(0.004)	0.02***	(0.004)
CASE COUNT	0.00*	(0.002)	0.00	(0.002)	−0.00	(0.001)	−0.00	(0.002)
S.STOCK	0.03***	(0.001)	0.02***	(0.002)	−0.01***	(0.002)	0.00	(0.003)
B.STOCK	−0.00	(0.000)	−0.00***	(0.000)	0.00***	(0.000)	0.00	(0.001)
T.STOCK	−0.00***	(0.000)	−0.00***	(0.000)	0.00	(0.000)	0.00	(0.000)
IND. RETAIL SALES	0.02***	(0.005)	0.04***	(0.007)	−0.00	(0.007)	0.03*	(0.011)
CPI	−0.37***	(0.016)	−0.99***	(0.026)	0.16***	(0.025)	−0.23***	(0.040)
GDP	0.00*	(0.002)	0.00	(0.003)	0.00*	(0.003)	0.01*	(0.005)
STORES	0.00	(0.005)	0.00	(0.007)	0.00	(0.005)	0.00	(0.005)
CLOSURE	0.04*	(0.020)	0.03	(0.031)	0.00	(0.027)	−0.01	(0.036)
STORE SALES	–	–	–	–	1.28***	(0.012)	0.88***	(0.012)
N	15,672		15,672		15,672		14,599	
R^2	0.63		0.58		0.62		0.52	

Note: Standard errors are reported in parentheses.
**p* < 0.05, ***p* < 0.01, ****p* < 0.001.

products in the U.S. stores is negative and significant ($\delta_3 = -0.10$, $p < 0.001$) relative to Canadian stores that are not affected by STS. A potential explanation for this finding arises from the channel switching of customers. After the STS rollout, when online customers elect the STS shipping option, they may learn that a store is located near them and decide that they prefer to make a purchase in-person at the BM store. Customers may also decide they would rather not wait up to three business days for store delivery. Either way, it is clear that customer movement to the BM channel occurs after STS customers conduct online research about the products, prices, assortment, and other product characteristics of the item(s) they are interested in purchasing. This activity typifies what is known as *reverse showrooming* or *webrooming*, in which customers gather information online, and after making a purchase decision, go to a physical store to make the actual purchase. Since these customers are more informed, they likely make better decisions, and consequently are less likely to return purchases.

5.2. Aggregate analysis

Even though we observe sales decline in one channel and increase in the other channel post-STS, it remains unclear what happens at an aggregate level. Based on results in Table 4, we know online channel sales decreased by 14% in affected DMAs after the STS introduction, relative to unaffected DMAs. We also know BM channel sales in the U.S. stores increased by 15% after the STS introduction relative to Canadian stores, which were not affected (Table 9). Note that online channel sales represent a small portion (7.5%) of total sales. Hence, the BM channel sales increase appears to dominate, increasing overall sales by about 13%.

Although this approach helps us evaluate the aggregate impact on the retailer's total sales, we also employ a DID approach to investigate the aggregate outcome. Our unit of analysis for the aggregate impact is state-month. The treatment group includes states located in the U.S. and the control group includes provinces in Canada. Our dependent variable is the log of total sales (both online and BM channels) at state *s* during month *t* (*TOTAL SALES$_{st}$*). The model specification for total sales is given by:

$$\log(\text{TOTAL SALES}_{st}) = \mu_s + \gamma_1 * \text{GROUP}_s + \gamma_2 * \text{POLICY}_t + \gamma_3 * \text{GROUP}_s * \text{POLICY}_t + \gamma_a * \text{CONTROLS}_{ast} + \varepsilon_{dt} \quad (6)$$

We observe that after the STS rollout, there is a positive and significant effect ($\gamma_3 = 0.16$, $p < 0.001$, $R^2 = 0.76$) on the retailer's total sales in the U.S. states relative to Canadian provinces. Hence, this result further supports our finding that implementing STS increased the retailer's total sales. The Appendix describes a robustness analysis to investigate the potential existence of different pre-intervention trends for the control and treatment groups, and shows they follow the same trend during the pre-STS period.

5.3. Cross-channel analysis of STS

We next conduct cross-channel analyses for average purchase price and return rate during both pre-STS and post-STS periods. First, we use within-channel two-sample t-tests. Subsequently, we conduct cross-

Table 11
Cross-channel analysis.

		BM Channel	Online Channel	*p*-value	STS	*p*-value
AVERAGE PRICE	Pre-STS	$405.11	$176.74	0.000		
	Post-STS	$407.05	$183.88	0.000	$176.57	0.000
	p-value	0.012	0.000			
RETURN RATE	Pre-STS	0.187	0.167	0.038		
	Post-STS	0.183	0.162	0.004	0.183	0.015
	p-value	0.491	0.585			

channel two-sample t-tests for both pre-STS and post-STS periods. Consistent with prior sections, we use 12 months before and 12 months after the STS rollout. For this analysis, we classify STS transactions separately in order to tease out differences in customer activity between regular online purchases and STS purchases. We report results in Table 11. The fifth column in Table 11 reports the p-value for a two-sample *t*-test between online and BM channels. The seventh column reports the p-value for a two-sample *t*-test between STS transactions and the online channel.

We first investigate the average purchase price across channels before and after STS. Note we do not have any STS transactions in the pre-STS period. We show the average purchase price in both the BM and online channels increases (i.e., $405.11 to $407.05; $176.74 to $183.88) after STS. Our results also show the average purchase price for STS transactions is lower than for the online channel at large (i.e., $176.57 vs. $183.88). These findings indicate that after the STS rollout, customers switch from online to the BM channel to purchase high-value items, while they switch from home delivery to STS to purchase low-value items.

Next, we analyze whether the return rate changes across channels after introduction of STS. We compute return rate as a ratio of total refunds to total sales for each month and channel. We find within-channel return rates do not change from pre-STS to post-STS ($p = 0.491$; $p = 0.585$). Interestingly, however, the results also show that the online channel return rate is significantly lower than the BM channel return rate ($p = 0.038$; $p = 0.004$), running counter to prior work (Rogers and Tibben-Lembke, 1999; Dunn, 2015; Winkler, 2016).

On the one hand, online transactions have a higher level of product fit uncertainty since customers are not able to touch and feel the product and hence, have considerably less information than that available when visiting a store. Higher fit uncertainty should correspond to higher return rates. On the other hand, from the perspective of expectation disconfirmation theory (Oliver, 1977), one would expect that the higher priced items purchased in stores (on average) will generate higher expectations regarding product quality since price is an indicator of quality (Zeithaml, 1988). Higher expectations lead to a higher probability that the disconfirmation of those pre-purchase beliefs will be negative, relative to lower-priced items. Any negative disconfirmation of the expected product performance translates into a higher level of product returns.

In the case of our present study, it appears that the effect of price trumps product fit uncertainty. Customers have more information when in stores, but return at a higher rate, ostensibly due to the higher price of products they purchase. With online purchases, customers have less information, but return at a lower rate, ostensibly due to the lower value of products they purchase. Of course, some other factors may be at play here, like the relative hassles of returning online versus offline. Nevertheless, the data provides an opportunity to more fully explore the uncertainties driving returns and the inherent tradeoffs with respect to price and fit uncertainty.

Table 12 presents return rates for both the BM channel and the online channel at different price bands. To create Table 12, we develop

Table 12
Return rates for price bands.

Channels	Low-Value	Medium-Value	High-Value	Max-Value
BM CHANNEL	7.87%	12.07%	14.93%	18.30%
ONLINE CHANNEL	7.53%	12.03%	16.26%	19.78%

four price bands: Low-Value (*Price* < $100), Medium-Value (100 ≤ *Price* < $300), High-Value (300 ≤ *Price* < $500), and Max-Value (*Price* ≥ $500). These results provide pretty clear evidence that items in an assortment that are offered at a higher price have return rates that are higher than those offered at lower prices. Note that this relationship demonstrates a between-product price effect. Prior research has shown a within-product price effect: the return rate for a given product increases with respect to price (Peterson and Kumar 2009). Our result here, however, are consistent with other studies that demonstrate low-value items are less likely to be returned than high-value items (Hess and Mayhew, 1997; Wang et al., 2009; De et al., 2013). Although, Rao et al. (2014) shows no relationship and in fact, Shang et al. (2018) reports the reverse: higher-priced items in an assortment have lower return rates on average. In the latter study, the authors propose that customers spend more time searching and gathering information pre-purchase for more expensive items, so that their post-purchase return rates are lower. Clearly, more research and more theory development is needed in this area to better understand even the most fundamental factors that influence customer return behavior.

As for product fit uncertainty, one would expect that for each price band, the online channel would have a higher return rate than the BM channel. But this is not what Table 12 shows. Evidently there is an interaction effect at work. Online return rates only exceed those of BM return rates for high-value and max-value items. These results enhance our contribution by illustrating relative influences of price and product fit uncertainty.

6. Conclusion

In this research, we explore how the introduction of STS service, allowing customers to buy items online and later pick them up at a nearby store, affects retail sales and customer returns. Our results are extensive and somewhat surprising, providing a fairly comprehensive analysis on the impact of STS service at a large national retailer. We find that STS decreased online sales, increased BM sales, increased cross channel returns, and reduced store returns. The fundamental understanding guiding management thinking about offering STS service is that STS should generate foot traffic at stores in the form of customer pick-ups and therefore provide secondary, additional selling opportunities for store employees. Indeed, our theory development initially led us to expect the same behavior. Yet, our results disconfirm this view at the jewelry retailer under study. The main effect of STS is not from generating ancillary store sales opportunities indirectly through online purchases. Rather, STS appears to directly increase store traffic and generate sales because some customers switch from the retailer's online channel to the BM channel to purchase high-value items. Moreover, STS draws new customers online that end up making their purchases in BM stores. The increase in BM store sales dwarfs the decrease we observe in online sales.

The direct increase in store sales due to STS is somewhat perplexing. The only direct impact that STS can make should be limited to the online channel, since a major inhibitor to online shopping (i.e., shipping fee) is removed via the free ship-to-store option. So, this message must be reaching online customers. However, instead of buying online and later picking up these purchases at a store, many potential STS customers go directly to the store to buy their items. Interestingly, this channel switching occurs mainly for high-value items. Furthermore, this customer movement occurs even though there is no store product availability information provided online, and many of the items that are available for purchase online are not even stocked in stores. One of the key promises offered to retailers for implementing STS is that it enables retailers to augment the physical inventory of their stores with a virtual inventory across the Internet. So, unless a customer calls ahead to the store prior to making a store visit, there is no way for the customer to know if the product is available at the store.

Another possible explanation is that the research customers conduct online makes them more comfortable with their own preferences and with the assortment offered by the retailer. In addition, since STS requires a visit to a store anyway, it probably makes sense for time-pressed customers (e.g., gift shoppers) to go directly to the store and avoid the three business day wait for STS delivery. Building on this latter argument is the notion of increasing the immediacy for purchase through successful search. That is, finding items online that they like, customers likely will then want those items more immediately (Kukar-Kinney and Close, 2010). The most immediate way to obtain the items is by immediately going to a local store.

Another interesting part of the story is that the increase in BM store sales is far more than a simple channel shift from online to BM. Our results indicate that the increase in BM store sales is far more than the decrease in online sales. Store customers are observed also to purchase higher-value items than online shoppers and buy more expensive items and more profitable services like product warranties. Consider that the average purchase for a store transaction was $405.11 prior to STS whereas for online customers it was less than half that, at $176.74. Yet the value of store customers after the STS rollout actually increases, with an average store purchase of $407.05. By all appearances, STS introduction generated BM store foot traffic of high-value store customers, but not low-value online customers.

Another outcome of STS service relates to its impact on product returns. We find that product returns of high-value items at the BM channel decrease, while overall returns for online purchases remain unchanged. As discussed in §5, the decrease in return rate for the BM channel likely arises because customers who switch from the online channel to the BM channel to purchase high-value items have conducted prior online research. As a result, they are probably more knowledgeable about the products and product assortment compared to other customers. In turn, more informed customers return less. The same argument also applies to consumer returns as well. Because customers are more knowledgeable about their own preferences along with a product's ability to meet their needs, they are able to make more informed decisions regarding a product's fit.

The only ancillary sales opportunities provided by STS are those that arise from (a) the small subset of customers buying low-value items that actually end up using the STS service, and also from (b) an increase in RTS incidents. By definition, returning an item to a store increases store traffic and presents additional selling opportunities. It is interesting to note that the average STS return rate is nearly identical to the average store return rate of 18.3%, which is substantially higher than the average online return rate of 16.2%. This outcome arises even though STS purchases are less expensive than other online purchases and are considerably less expensive than typical store purchases. We believe there are two potential explanations. First, STS makes returning an item simple and easy since a customer is already at the store to pick up the item. In effect, the return policy is more lenient since it eliminates all the hassles and costs typically associated with returning online purchases. We have argued that return rates for online items are lower than for store purchases because the items that are purchased online are less expensive. We have also cited other studies that have observed that return rates for less expensive items are lower than for more expensive items. Yet, in the case of our jewelry retailer, the lower return rate for online items may also be, at least partly, attributed to the greater inconvenience posed by making returns of online purchases as compared to making in-store returns. The second potential explanation for a higher return rate of STS purchases is that store sales associates are able to intervene at the point of pickup and steer customers to other higher-priced items. As our analysis shows, 28% of RTS incidents result in new sales.

The least valuable customers are the ones that ultimately end up using the STS service. The average STS purchase is $176.57. Overall, customers buy less expensive items online since there is less risk with these purchases. The low price customers presumably use the STS service since it provides the greatest relative perceived benefit, in terms of a disproportionate savings provided by free STS store shipping. Not only are these customers less valuable, but they are costlier since the retailer now absorbs STS shipping costs for these customers. With a decrease in online sales and an increase in the cost of servicing STS customers, the implementation of STS service might be viewed as a failure, if viewed solely from an online channel perspective. Such a singular view, however, would ignore the observed cross-channel impact on the BM channel and would run counter to the omnichannel retailing strategy that is enabled through STS.

As one of the earliest contributions on the impact of STS service on omnichannel retailing, there are, of course, plenty of opportunities for future research. A few promising avenues would include exploring the potential moderating impact of product category on STS. Here, we explore jewelry products, but these are luxury, highly experiential products in which search is quite important. Do the same effects hold for other product categories? We believe another significant opportunity lies in comparing and contrasting STS against BOPS and their hybrids. What are the operating conditions in which one of these services is preferred to another? For that matter, when should retailers avoid these services altogether? Finally, we think future research should explore the impact of lead time on omnichannel customer behavior. In our research, we cannot distinguish how lead time for store delivery affects customer switching behavior. Would a shorter or longer lead time have the same effect?

APPENDIX

Robustness Analysis

In this section, we report results of several robustness tests we conducted to further validate the findings. We organize the robustness runs according to the nature of robustness or the related section of the paper to which the results pertain.

Robustness to Changes in Dependent Variable Measurement First, we fully replicated the analyses for the BM channel and for the online channel by using variables measured using a dollar value as well as count variables for each variable of interest (i.e. purchases, returns, and returns to stores). Note that the findings we report in §4.2 and §4.3 are based on the dollar-value of variables. Our goal in this section is to report those same results by using the count equivalent of the dependent variables (e.g. number of purchases and returns). We report the results for online channel purchases in Table 13, online channel returns in Table 14, and BM channel purchases in Table 15. Observing the variable of interest, *GROUP∗POLICY*, across different models, we see that our findings do not change when we use the count variables for each variable of interest.

Table 13
Impact of STS on Online Channel Purchases (Count Variable).

Variables	Online pur.		High value pur.		Low value pur.	
GROUP	0.50∗∗∗	(0.077)	0.38∗∗∗	(0.073)	0.52∗∗∗	(0.084)
POLICY	0.23∗∗∗	(0.040)	0.04	(0.042)	0.27∗∗	(0.054)
GROUP*POLICY	−0.11∗∗∗	(0.030)	−0.10∗∗	(0.039)	−0.05	(0.042)
HOLIDAY	1.16∗∗∗	(0.021)	1.14∗∗∗	(0.022)	1.27∗∗∗	(0.027)
SUMMER	−0.30∗∗∗	(0.019)	−0.29∗∗∗	(0.020)	−0.37∗∗∗	(0.024)
PROMOTION	1.03∗∗∗	(0.047)	0.75∗∗∗	(0.045)	0.70∗∗∗	(0.046)
DMA RETAIL SALES	0.99∗∗∗	(0.022)	0.98∗∗∗	(0.021)	1.00∗∗∗	(0.022)
DMA UNEMPLOYMENT	0.02	(0.012)	0.02	(0.012)	0.02	(0.013)
S.STOCK	0.02∗∗∗	(0.003)	0.03∗∗∗	(0.003)	0.01∗	(0.003)
B.STOCK	0.02∗∗∗	(0.001)	0.01∗∗∗	(0.001)	0.02∗∗∗	(0.002)
T.STOCK	−0.00∗∗∗	(0.001)	−0.00∗∗∗	(0.001)	−0.00∗∗∗	(0.001)
IND. RETAIL SALES	0.10∗∗∗	(0.012)	0.06∗∗∗	(0.012)	0.17∗∗∗	(0.015)
CPI	−0.73∗∗∗	(0.041)	−0.70∗∗∗	(0.044)	−0.83∗∗∗	(0.054)
GDP	0.02∗∗∗	(0.004)	0.01∗	(0.005)	0.03∗∗∗	(0.006)
ESALES	−0.03∗∗∗	(0.007)	0.02∗	(0.007)	−0.07∗∗∗	(0.009)
N	5,024		5,009		4,966	
R^2	0.71		0.67		0.64	

Table 14
Impact of STS on Online Channel Returns (Count Variable).

Variables	Cross returns		Online returns	
GROUP	0.12	(0.106)	−0.22∗∗∗	(0.045)
POLICY	−0.26∗	(0.123)	0.02	(0.060)
GROUP*POLICY	0.31∗∗	(0.103)	0.03	(0.047)
ONLINE PURCHASES	1.04∗∗∗	(0.023)	0.85∗∗∗	(0.019)
HOLIDAY	−0.31∗∗∗	(0.060)	−0.01	(0.035)
SUMMER	0.08	(0.049)	0.17∗∗∗	(0.027)
PROMOTION	0.11	(0.076)	−0.35∗∗∗	(0.077)
DMA RETAIL SALES	–	–	0.16∗∗∗	(0.021)
UNEMPLOYMENT	0.05∗∗∗	(0.012)	−0.01∗	(0.006)
S.STOCK	0.00	(0.007)	−0.01∗∗	(0.004)
B.STOCK	−0.01∗∗	(0.004)	0.00	(0.002)
T.STOCK	−0.00	(0.002)	0.01∗∗∗	(0.001)
IND. RETAIL SALES	−0.02	(0.030)	0.06∗∗∗	(0.016)
CPI	0.24∗	(0.108)	−0.04	(0.059)
GDP	−0.02	(0.012)	−0.01	(0.006)
ESALES	0.02	(0.018)	0.00	(0.010)
N	1,160		4,667	
R^2	0.75		0.75	

Table 15
Impact of STS on Store Purchases (Count Variable).

Variables	Store purchases	
GROUP	−0.52∗∗∗	(0.030)
POLICY	−0.23∗∗∗	(0.019)
GROUP*POLICY	0.10∗∗∗	(0.018)
HOLIDAY	1.30∗∗∗	(0.010)
SUMMER	−0.41∗∗∗	(0.008)
PROMOTION	0.64∗∗∗	(0.041)
MALL GRADE A+	0.03	(0.034)
MALL GRADE B	−0.03	(0.021)

(*continued on next page*)

Table 15 (*continued*)

Variables	Store purchases	
MALL GRADE C	−0.03	(0.022)
MALL GRADE F	−0.04	(0.028)
MALL GRADE-COM	−0.04	(0.090)
MALL GRADE-LIF	−0.15***	(0.043)
MALL GRADE-MET	0.21*	(0.103)
MALL GRADE-POW	−0.09	(0.124)
MALL GRADE-VIL	0.15	(0.091)
VOLUME GROUP A+	0.21***	(0.046)
VOLUME GROUP B	−0.43***	(0.027)
VOLUME GROUP B+	−0.24***	(0.028)
VOLUME GROUP C	−0.67***	(0.029)
STORE SIZE	0.00***	(0.000)
INVENTORY TURNS	0.49***	(0.051)
UNEMPLOYMENT	−0.00	(0.003)
CASE COUNT	0.00*	(0.001)
S.STOCK	0.03***	(0.001)
B.STOCK	−0.00***	(0.000)
T.STOCK	−0.00***	(0.000)
IND. RETAIL SALES	0.03***	(0.005)
CPI	−0.75***	(0.017)
GDP	−0.00	(0.002)
STORES	0.00	(0.003)
CLOSURE	0.05	(0.021)
N	15,672	
R^2	0.70	

Robustness to Change in Treatment Group (Canada Brand) Second, we fully replicated the analyses for the impact of STS on the BM channel by using a different treatment group. We conduct this analysis for two reasons. First, observed differences might be explained by cultural or economic differences between the U.S. and Canada, rather than by the impact of the STS introduction. Second, a credit card program, which was implemented in the U.S. concurrently with the STS service, may confound results. To address these concerns, we employ an additional data set, which belongs to a different set of stores in Canada that the retailer owns and operates under a different brand name. We note that the retailer started implementing STS at this brand in September 2012, about a year after the U.S. stores. Thus, in this analysis we use 10 months before and 10 months after the STS introduction, since there are only 10 months in our data set after September 2012. Our data set for this analysis includes a total of 194 stores from Canada. Of those 194 stores, 143 were affected by STS and 51 were not affected. The control group is the same as the group that we used in §4.3 while the treatment group is different. We conduct the exact same analyses as explained in §4.3 by using these 194 stores. Table 16 shows that the coefficient for the variable of interest, *GROUP*POLICY*, for store sales is positive and significant ($p < 0.001$, $R^2 = 0.76$) with the point estimate being $\delta_3 = 0.20$. Hence, we conclude that this Canadian STS rollout increases BM store sales, corroborating our main findings.

Table 16
Different Treatment Group-1.

Variables	Store sales	
GROUP	−0.04	(0.032)
POLICY	−0.25***	(0.031)
GROUP*POLICY	0.20***	(0.018)
HOLIDAY	0.94***	(0.016)
SUMMER	−0.05*	(0.021)
PROMOTION	1.13***	(0.054)
MALL GRADE A+	−0.05	(0.055)
MALL GRADE B	−0.06	(0.036)
MALL GRADE C	−0.16**	(0.049)
MALL GRADE F	−0.09	(0.084)
VOLUME GROUP A+	0.06	(0.055)
VOLUME GROUP B	−0.50***	(0.058)
VOLUME GROUP B+	−0.23***	(0.054)
VOLUME GROUP C	−0.83***	(0.067)
STORE SIZE	0.00	(0.000)
INVENTORY TURNS	1.00***	(0.080)
UNEMPLOYMENT	−0.01	(0.007)

(*continued on next page*)

Table 16 (*continued*)

Variables	Store sales	
CASE COUNT	0.00	(0.003)
S.STOCK	0.00	(0.002)
B.STOCK	0.01***	(0.002)
T.STOCK	−0.01***	(0.002)
IND. RETAIL SALES	0.00	(0.007)
CPI	−0.15***	(0.017)
GDP	0.05***	(0.009)
STORES	−0.01	(0.009)
CLOSURE	−0.23	(0.058)
N	3,800	
R^2	0.76	

Robustness to Change in Treatment Group (Non-Flagship Brands) Third, we replicated the analyses for the impact of STS introduction on the BM channel by using the non-flagship store brands in the treatment group. By doing so, we address concerns that may arise due to potential economic differences between the flagship brand and non-flagship brands. Our data set for this analysis includes a total of 294 stores from both the U.S. and Canada. Of those 294 stores, 123 belong to the secondary brand and 123 belong to the U.S. outlet brand. STS was implemented for these 246 stores concurrently with the flagship brand stores. As explained in §4.3, the remaining 51 Canadian stores constitute the control group in our analysis. We conduct the exact same analyses as explained in §4.3 by using these 294 stores. Table 17 shows that the coefficient for the variable of interest, *GROUP*POLICY*, for store sales is positive and significant ($p < 0.005$, $R^2 = 0.60$) with the point estimate being $\delta_3 = 0.06$. Thus, our findings regarding the increase in BM store sales do not change.

Table 17
Different Treatment Group-2.

Variables	Store sales	
GROUP	−0.17***	(0.035)
POLICY	−0.15***	(0.021)
GROUP*POLICY	0.06**	(0.019)
HOLIDAY	0.98***	(0.013)
SUMMER	−0.19***	(0.012)
PROMOTION	0.77***	(0.057)
MALL GRADE A+	−0.05	(0.069)
MALL GRADE B	−0.06	(0.032)
MALL GRADE C	−0.13***	(0.034)
MALL GRADE F	0.07	(0.049)
MALL GRADE-COM	−0.23	(0.130)
MALL GRADE-LIF	−0.11	(0.174)
MALL GRADE-METRO	−0.08	(0.054)
MALL GRADE-POWER	−0.07	(0.050)
MALL GRADE-VILLAGE	−0.10*	(0.038)
VOLUME GROUP A+	0.06	(0.100)
VOLUME GROUP B	−0.60***	(0.040)
VOLUME GROUP B+	−0.28***	(0.042)
VOLUME GROUP C	−0.93***	(0.046)
STORE SIZE	0.00*	(0.000)
INVENTORY TURNS	0.43***	(0.076)
UNEMPLOYMENT	−0.00	(0.005)
CASE COUNT	0.01**	(0.002)
S.STOCK	0.02***	(0.001)
B.STOCK	−0.00**	(0.001)
T.STOCK	−0.00***	(0.001)
IND. RETAIL SALES	0.02***	(0.006)
CPI	−0.39***	(0.021)
GDP	0.01*	(0.003)
N	7,056	
R^2	0.60	

Robustness to Change in Control Group Fourth, we replicated the analyses for the impact of STS introduction on the BM channel by using a different control group. We conduct this analysis to address concerns arising from potential differences between brands. To do so, we employ the same data set from Canada, as described in the previous paragraph. Our data set for this analysis includes a total of 745 stores from both the U.S. and Canada. Of those 745 stores, 602 stores were affected by STS introduction and 143 stores were not affected. Note that the 143 Canadian stores were not affected by STS introduction between September 2011 and September 2012. We conduct the exact same analyses as explained in §4.3 by using these 745 stores. Table 18 shows that the coefficient for the variable of interest, *GROUP*POLICY*, for store sales is positive and significant ($p < 0.001$, $R^2 = 0.65$) with the point estimate being $\delta_3 = 0.06$. Hence, our findings regarding the increase in store sales are again corroborated.

Table 18
Different Control Group.

Variables	Store sales	
GROUP	−0.11***	(0.020)
POLICY	−0.09***	(0.012)
GROUP*POLICY	0.06***	(0.011)
HOLIDAY	1.04***	(0.008)
SUMMER	−0.19***	(0.007)
PROMOTION	0.79***	(0.035)
MALL GRADE A+	−0.02	(0.034)
MALL GRADE B	−0.04*	(0.021)
MALL GRADE C	−0.11***	(0.021)
MALL GRADE F	−0.19***	(0.027)
MALL GRADE-COM	−0.22*	(0.093)
MALL GRADE-LIF	−0.15***	(0.044)
MALL GRADE-METRO	0.11	(0.106)
MALL GRADE-POWER	−0.17	(0.105)
MALL GRADE-VILLAGE	−0.10	(0.093)
VOLUME GROUP A+	0.21***	(0.042)
VOLUME GROUP B	−0.63***	(0.026)
VOLUME GROUP B+	−0.32***	(0.026)
VOLUME GROUP C	−0.97***	(0.028)
STORE SIZE	0.00*	(0.000)
INVENTORY TURNS	0.60***	(0.046)
UNEMPLOYMENT	0.01	(0.003)
CASE COUNT	0.00***	(0.001)
S.STOCK	0.02***	(0.001)
B.STOCK	−0.00***	(0.001)
T.STOCK	−0.00***	(0.000)
IND. RETAIL SALES	0.02***	(0.004)
CPI	−0.40***	(0.013)
GDP	0.01**	(0.002)
N	17,880	
R^2	0.65	

Robustness to Change in Unit of Analysis Fifth, we replicated the analyses for the impact of STS on the online channel by using two different units of analysis. To validate our results in §4.2, we first use state-month as our unit of analysis rather than DMA-month. We conduct this analysis to dismiss concerns that may arise from the heterogeneity of DMAs. Table 19 shows that the coefficient for the variable of interest, *GROUP*POLICY*, for online sales is negative and significant ($p < 0.05$, $R^2 = 0.75$) with the point estimate being $\alpha_3 = -0.11$. We also find that the introduction of STS service does not have a statistically significant impact on online consumer returns ($p > 0.05$, $R^2 = 0.82$).

Table 19
Change in Unit of Analysis for the Online Channel-1.

Variables	Online sales		Online returns	
GROUP	1.56***	(0.406)	0.15	(0.079)
POLICY	0.17**	(0.063)	−0.10	(0.105)
GROUP*POLICY	−0.11*	(0.050)	0.12	(0.084)
ONLINE SALES	–	–	1.06***	(0.019)
HOLIDAY	0.98***	(0.031)	−0.20***	(0.054)
SUMMER	−0.22***	(0.027)	0.04	(0.046)
PROMOTION	1.49***	(0.099)	−0.43*	(0.166)

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Table 19 (*continued*)

Variables	Online sales		Online returns	
UNEMPLOYMENT	0.01	(0.022)	0.01	(0.009)
S.STOCK	0.02***	(0.004)	−0.01	(0.006)
B.STOCK	0.01***	(0.002)	−0.00	(0.003)
T.STOCK	−0.01***	(0.001)	0.01***	(0.002)
IND. RETAIL SALES	0.05**	(0.017)	0.04	(0.028)
CPI	−0.56***	(0.059)	0.11	(0.099)
GDP	−0.00	(0.006)	−0.02	(0.011)
ESALES	0.00	(0.010)	0.01	(0.017)
N	1,200		1,200	
R^2	0.75		0.82	

We also perform the same analyses by creating a treatment and a control group within each DMA. For this case, the control group consists of customers that are not within the influence area of a physical store (> 50 miles). The treatment group consists of customers within the influence area of a store. Once again, the results in Table 20 indicate that the coefficient for the variable of interest, *GROUP∗POLICY*, for online sales is negative and significant ($p < 0.001$, $R^2 = 0.30$) with the point estimate being $\alpha_3 = -0.10$. We also find that the introduction of STS does not have a statistically significant impact on online consumer returns ($p > 0.05$, $R^2 = 0.45$).

Table 20
Change in Unit of Analysis for the Online Channel-2.

Variables	Online sales		Online returns	
GROUP	2.71***	(0.096)	0.02	(0.052)
POLICY	0.05	(0.049)	0.05	(0.071)
GROUP*POLICY	−0.10***	(0.029)	0.05	(0.044)
ONLINE SALES	–	–	0.83***	(0.015)
HOLIDAY	1.05***	(0.031)	−0.09*	(0.043)
SUMMER	−0.30***	(0.028)	0.15***	(0.039)
PROMOTION	0.58***	(0.046)	−0.08	(0.090)
DMA RETAIL SALES	0.71***	(0.044)	0.13***	(0.019)
DMA UNEMPLOYMENT	−0.03	(0.027)	0.01	(0.011)
S.STOCK	0.04***	(0.004)	−0.01*	(0.005)
B.STOCK	0.01***	(0.002)	−0.00	(0.003)
T.STOCK	−0.01***	(0.001)	0.01***	(0.002)
IND. RETAIL SALES	0.10**	(0.018)	0.04	(0.024)
CPI	−0.75***	(0.063)	0.23**	(0.086)
GDP	0.01	(0.007)	−0.01	(0.009)
ESALES	−0.01	(0.011)	0.01	(0.014)
N	8,401		6,111	
R^2	0.30		0.45	

Robustness to Definition of High-Value Products Sixth, we conducted analyses for sales of high-value and low-value products both at the BM channel and the online channel by using product categories rather than a value threshold ($100). The product categories that we explore for these analyses include bridal, gold, and silver. Overall, these three categories represent about half of the total sales for the retailer. Products in the bridal category are the highest valued items in the data set. Table 21 shows that online sales of bridal items decrease while Table 22 shows that BM sales of bridal items increase. For the gold and silver categories, we observe no change at the BM channel or at the online channel. Hence, these findings further validate that, after STS is rolled out, customers switch from the online channel to the BM channel to purchase high-value products, while other customers continue using the online channel to purchase low-value products.

Table 21
Sales of Product Categories at Online Channel.

Variables	Bridal sales		Gold sales		Silver sales	
GROUP	0.17	(0.099)	0.37***	(0.088)	0.22*	(0.088)
POLICY	0.06	(0.093)	0.11	(0.068)	0.26***	(0.076)
GROUP*POLICY	−0.21**	(0.073)	−0.01	(0.053)	−0.05	(0.059)
HOLIDAY	0.79***	(0.045)	1.16***	(0.034)	1.43***	(0.037)
SUMMER	−0.38***	(0.041)	−0.43***	(0.032)	−0.31***	(0.034)
PROMOTION	0.52***	(0.065)	0.31***	(0.048)	0.46***	(0.050)

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Table 21 (*continued*)

Variables	Bridal sales		Gold sales		Silver sales	
DMA RETAIL SALES	0.93***	(0.027)	0.99***	(0.024)	0.99***	(0.024)
DMA UNEMPLOYMENT	0.01	(0.015)	0.01	(0.014)	0.01	(0.014)
S.STOCK	0.04***	(0.006)	0.04***	(0.004)	0.03***	(0.005)
B.STOCK	−0.01***	(0.003)	0.02**	(0.002)	0.02***	(0.003)
T.STOCK	−0.00*	(0.002)	−0.01***	(0.001)	−0.00**	(0.002)
IND. RETAIL SALES	0.07**	(0.026)	0.01	(0.019)	0.15***	(0.021)
CPI	−0.43***	(0.093)	−1.07***	(0.068)	−0.92***	(0.074)
GDP	−0.01	(0.010)	0.04***	(0.007)	0.02*	(0.008)
ESALES	0.02	(0.016)	−0.07	(0.012)	0.00	(0.013)
N	4,784		4,878		4,797	
R^2	0.28		0.53		0.54	

Table 22
Sales of Product Categories at BM Channel.

Variables	Bridal sales		Gold sales		Silver sales	
GROUP	0.30***	(0.040)	−0.79***	(0.050)	−0.43***	(0.051)
POLICY	−0.17***	(0.023)	−0.21***	(0.029)	−0.28***	(0.035)
GROUP*POLICY	0.15***	(0.021)	0.04	(0.027)	−0.05	(0.033)
HOLIDAY	0.67***	(0.011)	1.31***	(0.014)	1.65***	(0.017)
SUMMER	−0.11***	(0.010)	−0.34***	(0.013)	−0.55***	(0.015)
PROMOTION	0.85***	(0.034)	0.58***	(0.036)	0.92***	(0.065)
MALL GR. A+	−0.06	(0.047)	0.05	(0.059)	0.07	(0.052)
MALL GR. B	−0.03	(0.030)	−0.00	(0.037)	−0.06	(0.033)
MALL GR. C	−0.11***	(0.030)	−0.00	(0.038)	−0.06	(0.034)
MALL GR. F	−0.24***	(0.040)	0.05	(0.049)	−0.05	(0.045)
MALL GR.-COM	−0.39**	(0.127)	0.23	(0.158)	−0.10	(0.141)
MALL GR.-LIF	−0.09	(0.061)	−0.28***	(0.076)	−0.07	(0.069)
MALL GR.-MET	−0.01	(0.145)	0.25	(0.180)	0.42**	(0.160)
MALL GR.-POW	−0.05	(0.175)	−0.23	(0.217)	−0.09	(0.193)
MALL GR.-VIL	−0.33**	(0.128)	0.30	(0.159)	0.53***	(0.141)
VOL. GR. A+	0.28***	(0.065)	0.23**	(0.081)	0.19**	(0.072)
VOL. GR. B	−0.71***	(0.037)	−0.50***	(0.046)	−0.41***	(0.041)
VOL. GR. B+	−0.33***	(0.039)	−0.31***	(0.049)	−0.22***	(0.044)
VOL. GR. C	−1.09***	(0.041)	−0.73***	(0.050)	−0.61***	(0.045)
STORE SIZE	0.00	(0.000)	0.00***	(0.000)	0.00**	(0.000)
INVEN. TURNS	0.59***	(0.072)	0.56***	(0.090)	0.56***	(0.081)
UNEMP. RATE	−0.00	(0.005)	0.02**	(0.006)	0.01	(0.006)
CASE COUNT	0.00	(0.002)	0.01*	(0.003)	−0.00	(0.002)
S.STOCK	0.02***	(0.001)	0.02***	(0.002)	0.04***	(0.002)
B.STOCK	−0.00***	(0.000)	−0.00	(0.001)	0.00***	(0.001)
T.STOCK	−0.00***	(0.000)	−0.00***	(0.001)	−0.01***	(0.001)
IND. RETAIL SALES	0.03***	(0.006)	−0.00	(0.007)	0.04***	(0.009)
CPI	−0.24***	(0.020)	−0.52***	(0.026)	−1.06***	(0.031)
GDP	0.01**	(0.002)	0.01**	(0.003)	−0.00	(0.004)
STORES	0.00	(0.004)	0.00	(0.005)	0.01	(0.005)
CLOSURE	0.04	(0.025)	0.05	(0.032)	0.03	(0.037)
N	15,672		15,672		15,360	
R^2	0.40		0.50		0.55	

Robustness to Metric for High-Value Products Seventh, we explore whether warranty sales change across the online and BM channels after STS service is introduced. Warranty options available for customers include buying a lifetime protection plan or buying a lifetime protection plan with a two-year theft replacement. Consumers are more likely to purchase extended warranties for high-value products (Maronick, 2007). In this analysis, our dependent variables for the BM and online channels are the log of total warranty sales at store *j* during month *t* and at DMA *d* during month *t*, respectively. In Table 23, we see that, after STS was introduced to customers, online channel warranty sales decrease. In contrast, Table 24 shows that BM channel warranty sales increase after STS was introduced to customers. Hence, the results for warranty sales also support the finding of channel switching actions for high-value purchases from the online channel to the BM channel after STS introduction.

Table 23
Online Channel Warranty Sales.

Variables	Warranty sales	
GROUP	0.42***	(0.088)
POLICY	0.09	(0.063)
GROUP*POLICY	−0.32***	(0.047)
HOLIDAY	1.24***	(0.033)
SUMMER	−0.32***	(0.029)
PROMOTION	1.03***	(0.076)
UNEMPLOYMENT	0.01	(0.014)
S.STOCK	0.04***	(0.004)
B.STOCK	0.01***	(0.002)
T.STOCK	−0.01***	(0.001)
IND. RETAIL SALES	0.13**	(0.018)
CPI	−0.99***	(0.065)
GDP	−0.01	(0.007)
ESALES	−0.02	(0.011)
N	4,988	
R^2	0.54	

Table 24
BM Channel Warranty Sales.

Variables	Warranty sales	
GROUP	0.05	(0.038)
POLICY	−0.06**	(0.018)
GROUP*POLICY	0.10***	(0.017)
HOLIDAY	0.79***	(0.009)
SUMMER	−0.17***	(0.008)
PROMOTION	0.68***	(0.038)
MALL GRADE A+	−0.10*	(0.045)
MALL GRADE B	0.03	(0.028)
MALL GRADE C	0.02	(0.029)
MALL GRADE F	−0.04	(0.037)
MALL GRADE-COM	−0.07	(0.121)
MALL GRADE-LIF	−0.05	(0.058)
MALL GRADE-METRO	0.11	(0.138)
MALL GRADE-POWER	0.10	(0.166)
MALL GRADE-VILLAGE	−0.07	(0.121)
VOLUME GROUP A+	0.24***	(0.062)
VOLUME GROUP B	−0.63***	(0.035)
VOLUME GROUP B+	−0.33***	(0.037)
VOLUME GROUP C	−0.92***	(0.039)
STORE SIZE	0.00	(0.000)
INVENTORY TURNS	0.59***	(0.069)
UNEMPLOYMENT	0.01***	(0.004)
CASE COUNT	0.00	(0.002)
S.STOCK	0.02***	(0.001)
B.STOCK	−0.00***	(0.000)
T.STOCK	−0.00*	(0.000)

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Table 24 (*continued*)

Variables	Warranty sales	
IND. RETAIL SALES	0.02***	(0.004)
CPI	−0.31***	(0.016)
GDP	0.00	(0.002)
N	15,661	
R^2	0.52	

Robustness to Propensity Score Matching Analysis Eighth, we replicated our analyses for the BM and online channels by using propensity score matching. First, we report the results for the BM channel. The fact that our treatment group consists of 602 stores from the U.S. and control group consists of 51 stores from Canada may lead to some concerns related to selection bias and our goal with this additional analysis is to minimize those concerns. We first estimate the propensity scores for each store in our analysis using a logistic regression (logit) model. The dependent variable is of binary form, which takes a value of one if the store is in the treatment group and zero otherwise. The independent variables that we use in the logit model include MALL GRADE, VOLUME GROUP, STORE SIZE, INVENTORY TURNS, and TOTAL CASES. We use those store specific independent variables to be able to select almost identical stores both for the treatment and control groups. The results of our logit model show that the model fits well since the residual deviance is 275.65 while the logit model with only the intercept generates a deviance of 357.98. In the next step, we match the stores in the treatment and control groups by using the fitted values (propensity scores) from the logit model. Note that we only choose 51 stores from the treatment group since the number of stores in our control group is 51. For example, the mean INVENTORY TURNS for the treatment and control groups before matching are 0.895 and 0.829, respectively. After the matching, the mean INVENTORY TURNS for the treatment and control groups are 0.838 and 0.829, respectively. Similarly, mean TOTAL CASES for the treatment and control groups before matching are 33.023 and 30.137, respectively. After the matching, mean TOTAL CASES for the treatment and control groups are 30.176 and 30.137, respectively. These results show that matching treatment and control group stores via propensity scores helps us minimize the differences between the stores in the two groups. Once the matching process is completed, we run our DID model using only 102 stores (51 in each group) instead of the original 653 stores. Table 25 shows that the coefficient for the variable of interest, *GROUP*POLICY*, for store sales is positive and significant ($p < 0.001$, $R^2 = 0.66$) with the point estimate being $\delta_3 = 0.10$. Hence, propensity score matching analysis helps us minimize selection bias concerns for the BM channel. Note that our findings are similar for all the models in the BM channel analysis.

Table 25
Impact of STS on BM Channel.

Variables	Store sales	
GROUP	−0.12**	(0.044)
POLICY	−0.11***	(0.026)
GROUP*POLICY	0.10***	(0.026)
HOLIDAY	1.09***	(0.021)
SUMMER	−0.18***	(0.019)
PROMOTION	0.72***	(0.082)
MALL GRADE A+	0.06	(0.091)
MALL GRADE B	−0.06	(0.044)
MALL GRADE C	−0.11*	(0.051)
VOLUME GROUP B	−0.58***	(0.067)
VOLUME GROUP B+	−0.26***	(0.068)
VOLUME GROUP C	−0.97***	(0.073)
STORE SIZE	0.00	(0.000)
INVENTORY TURNS	0.27	(0.143)
UNEMPLOYMENT	−0.01	(0.009)
CASE COUNT	0.01	(0.004)
S.STOCK	0.01***	(0.023)
B.STOCK	−0.00*	(0.001)
T.STOCK	0.00	(0.001)
IND. RETAIL SALES	0.01	(0.009)
CPI	−0.32***	(0.027)
GDP	0.00	(0.005)
STORES	0.01	(0.009)
CLOSURE	0.01	(0.049)
N	2,448	
R^2	0.66	

Next, we repeat the same analysis for the online channel in order to minimize selection bias concerns for the treatment and control group DMAs. We first estimate the propensity scores for each DMA in our analysis using a logit model. The dependent variable is of binary form, which takes a value of one if the DMA is in the treatment group and zero otherwise. The independent variables that we use in the logit model include DMA UNEMPLOYMENT, AGE, HOME OWNER, INCOME, RESIDENCE, CHILD, and GENDER. We create AGE, INCOME, RESIDENCE variables for each DMA by taking the average of a categorical age status (13 levels) variable, a categorical income status (9 levels) variable, and a continuous length of residence (number of years) variable respectively, which are used by the retailer to classify its customers. Similarly, we create home owner, child, and gender variables by taking the average of a binary gender status, a binary child status, and a binary home owner status variables, respectively. The results of our logit model show that the model fits well since the residual deviance is 121.84 while the logit model with only the intercept generates a deviance of 153.31. For example, the mean income for the treatment and control groups before matching are 5.076 and 4.854, respectively. After the matching, the mean income for the treatment and control groups are 4.881 and 4.854. Similarly, mean DMA UNEMPLOYMENT for the treatment and control groups before matching are 5.012 and 4.564, respectively. After the matching, the mean DMA UNEMPLOYMENT for the treatment and control groups are 4.812 and 4.564, respectively. These results show that matching treatment and control group DMAs via propensity scores helps us minimize the differences between the DMAs in the two groups. Once the matching process is completed, we run our DID model using only 50 DMAs (25 in each group) instead of the original 210 DMAs. Table 26 shows that the coefficient for the variable of interest, *GROUP∗POLICY*, for online sales is negative and significant ($p < 0.05$, $R^2 = 0.51$) with the point estimate being $\delta_3 = -0.14$. Conducting the same analysis for cross returns, we find in Table 27 that the variable of interest, *GROUP∗POLICY*, is positive and significant ($p < 0.01$, $R^2 = 0.48$) with the point estimate being $\delta_3 = 0.70$. Hence, propensity score matching analyses help us minimize selection bias concerns for the online channel.

Table 26
Impact of STS on Online Channel Sales.

Variables	Online sales		High value sales		Low value sales	
GROUP	0.51∗∗∗	(0.123)	0.48∗∗∗	(0.102)	0.59∗∗∗	(0.105)
POLICY	0.15	(0.107)	0.17	(0.111)	0.06	(0.112)
GROUP*POLICY	−0.14∗	(0.065)	−0.14∗	(0.068)	−0.05	(0.068)
HOLIDAY	1.21∗∗∗	(0.071)	1.16∗∗∗	(0.074)	1.32∗∗∗	(0.071)
SUMMER	−0.30∗∗∗	(0.064)	−0.23∗∗∗	(0.067)	−0.47∗∗∗	(0.064)
PROMOTION	0.57∗∗∗	(0.122)	0.73∗∗∗	(0.109)	0.66∗∗∗	(0.046)
DMA RETAIL SALES	0.96∗∗∗	(0.057)	0.94∗∗∗	(0.046)	0.90∗∗∗	(0.048)
DMA UNEMPLOYMENT	−0.00	(0.038)	−0.01	(0.031)	−0.03	(0.031)
S.STOCK	0.04∗∗∗	(0.009)	0.03∗∗∗	(0.009)	0.03∗∗∗	(0.009)
B.STOCK	0.01∗	(0.005)	0.01∗	(0.005)	0.01∗∗	(0.005)
T.STOCK	−0.01∗∗	(0.003)	−0.01∗∗	(0.003)	−0.01∗∗	(0.003)
IND. RETAIL SALES	0.09∗	(0.040)	0.07	(0.042)	0.11∗∗	(0.041)
CPI	−0.75∗∗∗	(0.142)	−0.66∗∗∗	(0.148)	−0.75∗∗∗	(0.144)
GDP	0.00	(0.015)	0.00	(0.016)	0.04∗	(0.016)
ESALES	−0.01	(0.024)	−0.00	(0.025)	−0.04	(0.025)
N	1,185		1,173		1,134	
R^2	0.51		0.51		0.56	

Table 27
Impact of STS on Online Channel Returns.

Variables	Cross returns		Online returns	
GROUP	0.33	(0.501)	0.29	(0.193)
POLICY	−0.04	(0.396)	0.18	(0.356)
GROUP*POLICY	0.70∗∗	(0.245)	−0.16	(0.217)
ONLINE SALES	0.89∗∗∗	(0.177)	1.39∗∗∗	(0.087)
HOLIDAY	−0.00	(0.317)	−0.13	(0.257)
SUMMER	0.28	(0.248)	0.18	(0.213)
PROMOTION	0.88∗∗	(0.265)	−0.47	(0.404)
DMA RETAIL SALES	–	–	0.33∗∗	(0.112)
UNEMPLOYMENT	0.10	(0.119)	0.04	(0.050)
S.STOCK	−0.01	(0.033)	0.02∗∗	(0.030)
B.STOCK	0.01	(0.018)	0.01	(0.016)
T.STOCK	−0.02	(0.012)	−0.01	(0.010)
IND. RETAIL SALES	−0.10	(0.147)	0.03	(0.133)
CPI	−0.20	(0.519)	0.98∗	(0.475)
GDP	−0.04	(0.057)	0.04	(0.051)

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Table 27 (*continued*)

Variables	Cross returns		Online returns	
ESALES	−0.02	(0.086)	0.07	(0.080)
N	167		1,185	
R^2	0.48		0.52	

Robustness to Using Return Rate as Dependent Variable Finally, we conducted robustness analyses to investigate the impact of STS on returns by using return probability as the dependent variable rather than the dollar amount of returns. For ONLINE RETURNS and BM channel HIGH-LOW RETURNS, we divide the total amount of returns by the total amount of sales to calculate the new dependent variable *return rate*. For CROSS RETURNS, we divide the amount of online sales returned to stores by the total amount of returns to calculate the new dependent variable *return-to-store rate*. Before we present the results of the fractional response models with panel data, we first want to briefly discuss our findings for the DID model that we introduced in the manuscript. Although the linear DID model does not ensure that the fitted response value will be between zero and one, one can still obtain the magnitude of the effect by observing the variable of interest (Papke and Wooldridge, 2008). For online returns, we find that the coefficient for the variable of interest *GROUP∗POLICY* is not statistically significant ($p > 0.1$, $R^2 = 0.21$). For CROSS RETURNS, our results indicate that the coefficient for the variable of interest *GROUP∗POLICY* is positive and statistically significant ($p < 0.01$, $R^2 = 0.17$). As for the returns at the BM channel, we find that the coefficient for *GROUP ∗POLICY* is negative and statistically significant for HIGH RETURNS ($p < 0.001$, $R^2 = 0.23$) while it is not statistically significant for LOW RETURNS ($p > 0.1$, $R^2 = 0.40$). As such, the results of the robustness analyses with fractional response variables are consistent with the findings that we report in the manuscript.

We conduct these robustness analyses using methodologies that are developed to address fractional response variables in a longitudinal setting. Papke and Wooldridge (2008) and Ramalho et al. (2018) propose robust nonlinear (i.e. logit and probit) panel data models that can accommodate fractional response variables such as proportions. Although both methods are quite similar with respect to handling the fractional response variable, we follow Ramalho et al. (2018) to be able to observe the time-invariant covariates. The estimator proposed by Ramalho et al. (2018) uses cluster-robust standard errors and generates unbiased estimates for random effects. Note that this method can only use numerical covariates. Hence, we remove qualitative independent variables such as MALL GRADE and VOLUME GROUP from this analysis.

Table 28 reports the results of the fractional logit regression models for the online channel. We find that the coefficient for the variable of interest, *GROUP∗POLICY*, for CROSS RETURNS is positive and significant ($p < 0.05$) with the point estimate being $\delta_3 = 0.77$. Conducting the same analysis for ONLINE RETURNS, we observe that the coefficient for *GROUP∗POLICY* is not statistically significant.

Table 28
Fractional LOGIT Models for the Online Channel.

Variables	Cross returns		Online returns	
GROUP	−0.83∗∗∗	(0.296)	−0.38	(0.125)
POLICY	−0.82∗	(0.441)	0.27∗∗∗	(0.226)
GROUP*POLICY	0.77∗	(0.448)	−0.20	(0.185)
HOLIDAY	−0.11	(0.096)	−0.78∗∗∗	(0.070)
SUMMER	−0.10	(0.085)	0.27∗∗∗	(0.060)
PROMOTION	0.28∗	(0.167)	−0.95∗∗∗	(0.296)
DMA RETAIL SALES	–	–	−0.56∗∗∗	(0.064)
UNEMPLOYMENT	0.01	(0.027)	0.00	(0.010)
S.STOCK	0.01	(0.019)	−0.02∗∗	(0.008)
B.STOCK	−0.02∗∗∗	(0.007)	−0.01∗	(0.005)
T.STOCK	−0.01	(0.008)	0.01∗∗	(0.003)
IND. RETAIL SALES	−0.17∗∗∗	(0.059)	−0.05	(0.042)
CPI	0.56∗∗	(0.240)	0.83∗∗∗	(0.126)
GDP	0.00	(0.036)	0.00	(0.014)
NUMBER OF RETURNS	0.10∗∗	(0.046)	0.48∗∗∗	(0.057)
ESALES	0.04	(0.035)	−0.01	(0.022)
N	1,159		4,665	

Table 29 reports the results of the fractional logit regression models for the BM channel. We find that the coefficient for the variable of interest, *GROUP*POLICY*, for HIGH RETURNS is negative and significant ($p < 0.001$) with the point estimate being $\delta 3 = -0.11$. Conducting the same analysis for low returns, we observe that the coefficient for *GROUP*POLICY* is positive and significant ($p < 0.01$) with the point estimate being $\delta 3 = 0.06$. Note that the results for the fractional logit models are consistent with the findings that we report in the manuscript except for low returns at the BM channel. One potential explanation for low returns could be related to the fact that STS is used by customers primarily for low-value items. Once these customers visit stores to pick up their STS items, they could be purchasing additional low-value products through compulsive buying behavior. Since the mismatch probability is higher for these additional products, customers may be returning more.

Table 29
Fractional LOGIT Models for the BM Channel. Analysis of Preintervention Trends.

Variables	High returns		Low returns	
GROUP	0.24***	(0.045)	0.53***	(0.033)
POLICY	0.27***	(0.032)	0.08**	(0.033)
GROUP*POLICY	−0.11***	(0.029)	0.06**	(0.031)
HOLIDAY	−0.79***	(0.019)	−0.94***	(0.016)
SUMMER	0.14***	(0.015)	0.43***	(0.018)
PROMOTION	−0.63***	(0.123)	0.38***	(0.049)
STORE SIZE	−0.00***	(0.000)	−0.00***	(0.000)
INVEN. TURNS	−0.75***	(0.052)	0.54***	(0.062)
UNEMP. RATE	0.02***	(0.004)	0.03***	(0.007)
CASE COUNT	−0.00**	(0.002)	−0.00	(0.002)
S.STOCK	−0.02***	(0.002)	−0.00	(0.002)
B.STOCK	0.00*	(0.001)	0.00***	(0.001)
T.STOCK	0.00***	(0.001)	0.00	(0.001)
IND. RETAIL SALES	−0.00	(0.008)	−0.03***	(0.009)
CPI	0.35***	(0.031)	0.45***	(0.027)
GDP	0.00	(0.003)	−0.00	(0.004)
STORES	−0.01***	(0.004)	−0.00	(0.005)
CLOSURE	−0.03	(0.029)	−0.01	(0.039)
N	15,672		14,599	

Following (Gallino and Moreno, 2014), we examine whether control and treatment groups in our models follow the same trend during the preintervention (i.e., pre-STS) period. If the pre-STS trends for the control and treatment groups differ, we may still find an effect with the DID analysis, but unfortunately one that is arising from the difference in preintervention trends. In such a case, the effect that we find would be confounded with the pre-STS trends. To rule out any concerns about different pre-STS trends for the control and treatment groups in the online channel, we use the following model specification:

$$\log(\text{ONLINE SALES}_{dt}) = \mu_d + \alpha_1 * \text{GROUP}_d + \alpha_2 * \text{TREND}_t + \alpha_3 * \text{GROUP}_d * \text{TREND}_t + \alpha_m * CONTROLS_{mdt} + \varepsilon_{dt} \quad (7)$$

$TREND_t$ identifies the number of days since the beginning of the analysis period (September 2010). Table 30 shows that the coefficient for the variable of interest, *GROUP*TREND*, for online sales is not statistically significant ($p > 0.1$, $R^2 = 0.68$). Note that the same finding also applies to high-value and low-value sales. Thus, we conclude that the control and treatment groups follow the same trend in the online channel before STS service is implemented and rule out concerns that different pre-STS trends may be driving the results observed for the model in equation (1).

Table 30
Online Channel Preintervention Trends-1.

Variables	Online sales		High value sales		Low value sales	
GROUP	0.42**	(0.161)	0.37*	(0.168)	0.74***	(0.170)
TREND	−0.01***	(0.001)	−0.01	(0.001)	−0.01***	(0.001)
GROUP*TREND	0.00	(0.000)	0.00	(0.000)	−0.00	(0.000)
HOLIDAY	0.17***	(0.083)	1.75***	(0.091)	1.48***	(0.080)
SUMMER	−1.13***	(0.075)	−1.11***	(0.082)	−1.15***	(0.072)
PROMOTION	0.49**	(0.082)	0.54***	(0.076)	0.38***	(0.060)
DMA RETAIL SALES	0.79***	(0.188)	0.83***	(0.181)	0.82***	(0.175)
DMA UNEMPLOYMENT	0.02	(0.016)	0.03	(0.015)	0.00	(0.014)
S.STOCK	−0.09***	(0.014)	−0.09***	(0.015)	−0.06***	(0.013)
B.STOCK	−0.01***	(0.004)	−0.02***	(0.004)	0.00	(0.004)
T.STOCK	0.17***	(0.017)	0.18***	(0.019)	0.13***	(0.017)
IND. RETAIL SALES	0.53***	(0.076)	0.59***	(0.082)	0.23**	(0.073)

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Table 30 (*continued*)

Variables	Online sales		High value sales		Low value sales	
CPI	−0.07	(0.152)	0.10	(0.165)	−0.67***	(0.147)
GDP	−0.28***	(0.025)	−0.29***	(0.027)	−0.20***	(0.024)
ESALES	0.50***	(0.074)	0.56***	(0.081)	0.12	(0.072)
N	2,511		2,500		2,475	
R^2	0.68		0.65		0.74	

Conducting the same analysis for *CROSS RETURNS*, we see in Table 31 that the coefficient for the variables of interest, *GROUP∗TREND*, is not statistically significant ($p > 0.1$, $R^2 = 0.69$). Hence, we conclude that control and treatment groups follow the same trend during the preintervention period.

Table 31
Online Channel Preintervention Trends-2.

Variables	Cross returns	
GROUP	−0.50	(0.725)
TREND	−0.00	(0.002)
GROUP*TREND	0.00	(0.001)
ONLINE SALES	1.12***	(0.041)
HOLIDAY	−0.30	(0.302)
SUMMER	0.38	(0.268)
PROMOTION	0.54***	(0.157)
UNEMPLOYMENT	0.05**	(0.018)
S.STOCK	−0.02	(0.049)
B.STOCK	−0.01	(0.014)
T.STOCK	0.00	(0.062)
IND. RETAIL SALES	0.06	(0.269)
CPI	1.28*	(0.545)
GDP	−0.02	(0.089)
ESALES	0.10	(0.266)
N	575	
R^2	0.69	

Similar to the analysis we did for the online channel, we use the following model specification to rule out concerns about different pre-STS trends for the control and treatment groups in the BM channel.

$$\log(\text{STORE SALES}_{jt}) = \mu_j + \delta_1 * \text{GROUP}_j + \delta_2 * \text{TREND}_t + \delta_3 * \text{GROUP}_j * \text{TREND}_t + \delta_s * CONTROLS_{sjt} + \varepsilon_{dt} \tag{8}$$

$TREND_t$ counts the number of days since the beginning of the analysis period (September 2010). Table 32 shows that the coefficient for the variable of interest, *GROUP∗TREND*, for store sales is not statistically significant ($p > 0.1$, $R^2 = 0.72$). Therefore, we conclude that the control and treatment groups in the BM channel follow the same trend before STS service is introduced.

Finally, we conducted a similar analysis to investigate the pre-STS trends for the aggregate impact in §5.2. Table 33 shows that the coefficient for the variable of interest, *GROUP∗TREND*, for aggregate sales is not statistically significant ($p > 0.05$, $R^2 = 0.86$). Hence, we conclude that the retailer's total sales in the U.S. and Canada followed the same trend during the pre-STS period.

Table 32
BM Channel Preintervention Trends.

Variables	Store sales	
GROUP	−0.21**	(0.066)
TREND	−0.00***	(0.000)
GROUP*TREND	−0.00	(0.000)
HOLIDAY	1.01***	(0.012)
SUMMER	−0.42***	(0.014)
PROMOTION	1.19***	(0.064)
MALL GRADE A+	−0.02	(0.037)
MALL GRADE B	−0.06*	(0.024)
MALL GRADE C	−0.12***	(0.025)
MALL GRADE F	−0.22***	(0.032)

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Table 32 (*continued*)

Variables	Store sales	
MALL GRADE-COM	−0.27**	(0.100)
MALL GRADE-LIF	−0.15**	(0.048)
MALL GRADE-METRO	−0.01	(0.114)
MALL GRADE-POWER	−0.16	(0.137)
MALL GRADE-VILLAGE	−0.16	(0.101)
VOLUME GROUP A+	−0.29***	(0.051)
VOLUME GROUP B	−0.67***	(0.029)
VOLUME GROUP B+	−0.35***	(0.031)
VOLUME GROUP C	−1.01***	(0.032)
STORE SIZE	0.00	(0.000)
INVENTORY TURNS	0.29***	(0.057)
UNEMPLOYMENT	−0.00	(0.004)
CASE COUNT	0.00**	(0.002)
S.STOCK	0.00	(0.002)
B.STOCK	−0.00	(0.001)
T.STOCK	0.04***	(0.001)
IND. RETAIL SALES	−0.05***	(0.009)
CPI	−0.38***	(0.026)
GDP	−0.06	(0.003)
STORES	−0.01	(0.005)
CLOSURE	0.04	(0.028)
N	7,836	
R^2	0.72	

Table 33
Aggregate Analysis.

Variables	Aggregate sales	
GROUP	0.53	(0.458)
TREND	−0.00***	(0.000)
GROUP*TREND	0.00	(0.000)
HOLIDAY	1.01***	(0.025)
SUMMER	−0.35***	(0.029)
PROMOTION	2.20***	(0.170)
S.STOCK	−0.00	(0.004)
B.STOCK	−0.00*	(0.002)
T.STOCK	0.04***	(0.003)
IND. RETAIL SALES	−0.02	(0.018)
CPI	−0.30***	(0.051)
GDP	−0.08***	(0.005)
N	684	
R^2	0.86	

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